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A trimodal protein language model enables advanced protein searches

In the format provided by the authors and unedited

TrEMBL50							
Sequence-level				Residue-level			
Subsection	Num(F)	Num(A)	Num(F) / Num(A) (%)	Subsection	Num(F)	Num(A)	Num(F) / Num(A) (%)
Gene names	403362	930060	43.37	Domain	2929458	5885898	49.77
Protein names	57883	3979530	1.45	Chain	585920	1769394	33.11
Organism	20076	372085	5.40	Compositional bias	143116	801190	17.86
GO annotation	11371	3621371	0.31	Region	102923	1512529	6.80
Sequence similarities	6861	2145953	0.32	Repeat	75579	700456	10.79
Function	4405	509670	0.86	Binding site	61385	584807	10.50
Catalytic activity	2672	617558	0.43	Coiled coil	41573	154788	26.86
Proteomes	1673	12260	13.65	DNA binding	19364	242336	7.99
Subunit	1297	124955	1.04	Transmembrane	11971	840833	1.42
Pathway	726	161197	0.45	Active site	9607	124509	7.72
Taxonomic lineage	652	39964	1.63	Disulfide bond	8434	33640	25.07
Subcellular location	468	92646	0.51	Site	7064	39743	17.77
Cofactor	296	99807	0.30	Motif	5309	35961	14.76
Domain (non-positional annotation)	248	38189	0.65	Modified residue	4257	125752	3.39
Post-translational modification	119	9353	1.27	Topological domain	882	1508	58.49
Miscellaneous	61	8654	0.70	Cross-link	238	290	82.07
Activity regulation	45	4462	1.01	Lipidation	130	409	31.78
Caution	14	19553	0.07	Signal peptide	36	58736	0.06
Induction	9	129	6.98	Propeptide	33	73	45.21
				Intramembrane	8	10	80.00
				Peptide	7	19	36.84
				Glycosylation	4	8	50.00
Total	512,238	12,787,396		Total	4,007,298	12,912,889	

Swiss-Prot							
Sequence-level				Residue-level			
Subsection	Num(F)	Num(A)	Num(F) / Num(A) (%)	Subsection	Num(F)	Num(A)	Num(F) / Num(A) (%)
Gene names	666,632	1,121,127	59.46	Chain	252,675	575,915	43.87
Protein names	301,073	1,476,325	20.39	Region	119,507	319,532	37.40
Function	98,147	487,765	20.12	Binding site	108,215	1,186,207	9.12
Subunit	44,603	295,608	15.09	Domain	107,195	214,158	50.05
Tissue specificity	31,165	50,620	61.57	Mutagenesis	89,827	90,626	99.12
GO annotation	29,283	3,165,677	0.93	Compositional bias	75,066	174,232	43.08
Organism	26,684	881,200	3.03	Repeat	68,638	108,887	63.04
Subcellular location	19,972	509,901	3.92	Natural variant	64,392	64,820	99.34
Disruption phenotype	17,298	20,051	86.27	Topological domain	62,960	149,000	42.26
Post-translational modification	17,291	63,799	27.10	Disulfide bond	41,989	134,091	31.31
Miscellaneous	14,926	45,515	32.79	Modified residue	32,358	259,509	12.47
Catalytic activity	14,509	335,116	4.33	Transmembrane	24,967	380,879	6.56
Induction	14,210	25,198	56.39	Motif	17,405	47,202	36.87
Sequence similarities	14,161	518,327	2.73	Site	16,906	64,740	26.11
Developmental stage	11,576	14,119	81.99	Coiled coil	15,720	22,423	70.11
Domain (non-positional annotation)	10,159	57,737	17.60	Active site	12,870	176,303	7.30
Activity regulation	7,498	18,196	41.21	Glycosylation	9,309	123,475	7.54
Involvement in disease	6,359	6,845	92.90	Peptide	9,183	12,462	73.69
Taxonomic lineage	5,714	569,793	1.00	Propeptide	5,991	14,843	40.36
Caution	5,163	14,333	36.02	Cross-link	5,538	24,680	22.44
Biophysicochemical properties	3,398	7,835	43.37	DNA binding	4,465	12,152	36.74
Cofactor	2,017	230,702	0.87	Lipidation	2,172	13,723	15.83
Pathway	1,454	143,674	1.01	Intramembrane	1,254	2,969	42.24
Polymorphism	1,246	1,481	84.13	Transit peptide	383	8,162	4.69
RNA Editing	882	3,254	27.11	Signal peptide	185	43,855	0.42
Biotechnology	875	1,823	48.00				
Toxic dose	712	851	83.67				
Proteomes	600	490,045	0.12				
Allergenic properties	284	947	29.99				
Pharmaceutical use	154	165	93.33				
Total	1,368,045	10,558,029		Total	1,149,170	4,224,845	

Supplementary Table 1: Statistics of the pre-training dataset. **Num(F)** represents the number of unique text descriptions, while **Num(A)** indicates the total number of descriptions (including duplicate text) , that is, the number of protein-text pairs.

Sequence-level		Residue-level	
Subsection	**Example**	**Subsection**	**Example**
Gene names	The nucleotide sequence rpsI corresponds to the synthesis of this protein.	Chain	The protein's polypeptide chain at position 1 to 294 is recognized as Diaminopimelate epimerase.
Protein names	This protein is identified as Small ribosomal subunit protein uS9.	Region	This protein is labeled with Interaction with BTRC when in the region at position 1 to 151.
Function	Regulates a signal transduction pathway linking plasma membrane receptors to the assembly of focal adhesions and actin stress fibers.	Binding site	The residue at position 460 forms a bond with Mg(2+), indicating a strong binding affinity.
Subunit	Heterooligomer composed of large and small subunits.	Domain	At position 214 to 375, the domain is characterized as TrmE-type G.
Tissue specificity	Mostly expressed in stems and, to a lower extent, in roots and leaves.	Mutagenesis	The effect of altering the amino acid in this protein from I to A at position 582 is decreases kinase activity.
GO annotation	The GO term for this protein involving biological process incorporates leaf senescence.	Compositional bias	At position 299 to 314, the predominating amino acid type is Polar residues.
Organism	The organism's name Finless sole is linked to the protein sequence.	Repeat	The repeated sequence motif or domain at position 29 to 127 falls into the classification Solcar.
Subcellular location	The protein can be spotted at Cellular thylakoid lumen.	Natural variant	The protein's natural variant, known as in RNA edited version; does not alter GABA receptor activity;
Disruption phenotype	Reduced mating efficiency. Defective sperm. Abnormal distribution of pkd-2.	Topological domain	The protein's non-membrane segment is located at Cytoplasmic when considering its position 482 to 505.
Post-translational modification	Can be phosphorylated to form H3S10ph, H3T11ph and H3S28ph.	Disulfide bond	Cysteine residues are crucial in the formation of disulfide bonds at positions 113 and 341.
Miscellaneous	Present with 799 molecules/cell in log phase SD medium.	Modified residue	Phosphoserine designates the occurrence of a modified residue type at position 657 in this protein.
Catalytic activity	The reaction: ATP + H(+) + nicotinate beta-D-ribonucleotide = deamido-NAD(+) + diphosphate undergoes catalysis due to the presence of this protein.	Transmembrane	The protein's membrane-spanning segment at position 291 to 311 is identified as Helical.
Induction	Up-regulated by karrikins and strigolactone treatments.	Motif	The sequence motif 'HIGH' region occurs at the specific position 7 to 17 in this protein.
Sequence similarities	Belongs to the eukaryotic GSH synthase family.	Site	The residue found at 44 possesses the attribute of interacts with H-NS.
Developmental stage	Expressed during the parasite blood stage, in schizonts (at protein level).	Coiled coil	Position 115 to 135 in this protein is characterized by a coiled coil.
Domain (non-positional annotation)	The first C2 domain mediates Ca(2+)-dependent phospholipid binding.	Active site	Catalysis involves the residue at position 102 playing the role of Phosphoserine intermediate.
Activity regulation	Inhibited by C/VIF1 and C/VIF2.	Glycosylation	Position 230 hosts a glycan group that is covalently connected, and its type is N-linked (GlcNAc...) asparagine.
Involvement in disease	Isovaleric acidemia (IVA) is the disease tied to genetic variations in this protein.	Peptide	The active peptide at position 1 to 34 in this protein goes by the name Subtilosin-A.
Taxonomic lineage	In the taxonomic hierarchy, the source organism of this protein is classified within Bacteria.	Propeptide	The protein has been described to include a propeptide at position 181 to 203, which can be identified as Removed in mature form.
Caution	In strain K12 / CR63, finO is interrupted by an IS3 element.	Cross-link	The residues at positions 240 and 244, specifically H and Y, respectively, take part in an intrachain covalent linkage referred to as 1'-histidyl-3'-tyrosine (His-Tyr).
Biophysicochemical properties	Optimum pH is 8.0. The pH has no influence on the tropine to pseudotropine product ratio.	DNA binding	The DNA-binding domain occurring at position 265 to 340 is nuclear receptor.
Cofactor	Without pyruvate acting as a cofactor, this protein's function is compromised.	Lipidation	S-palmitoyl cysteine is the specific lipid group that forms a covalent attachment at position 19.
Pathway	The metabolic pathway that this protein is associated with consists of Carbohydrate biosynthesis; Calvin cycle.	Intramembrane	At position 121 to 127 in Pore-forming, one can observe the presence of the region embedded within a membrane.
Polymorphism	There are several alleles. The sequence shown is that of IMGT allele TRAV41*01.	Transit peptide	Through the transit peptide at 1 to 38, the protein is guided to Mitochondrion.
RNA Editing	This protein experiences RNA editing specifically at 494.	Signal peptide	The position 1 to 26 in this protein contains an N-terminal signal peptide.
Biotechnology	Could be a valuable biocatalyst used in the industrial field for an efficient biotransformation of mevastatin.		
Toxic dose	LD(50) is 0.06 ug/kg by intracerebroventricular injection into mice		
Proteomes	A proteome, containing Partially assembled WGS sequence, is composed of this protein.		
Allergenic properties	Causes an allergic reaction in human. Binds to IgE.		
Pharmaceutical use	Available under the name BeneFix (Baxter and American Home Products). Used to treat hemophilia B.		

Supplementary Table 2: Examples of text descriptions for each subsection. The examples in the residue-level subsections are created by filling in the keywords (e.g. position) in the templates.

Retrieval type	Corresponding task	Matching criteria
Protein-to-text	Fig. 2a	Texts are mapped to unique IDs. Positives and negatives are based on ID matching (see Methods 1.6).
	Supplementary Table 5	Manual examination of the predicted text description and core functions described in literature (keyword match serves as a prerequisite) (see Methods 1.16).
	Extended Data Fig. 2	The identification of CRISPR-associated proteins is based on the presence of CRISPR-Cas-related keywords. For enzymes, matches are evaluated through hierarchical alignment of EC numbers, with recognition of matches at multiple digit levels (see Methods 1.19).
Text-to-protein	Fig. 2a	Proteins are mapped to unique IDs. The target protein is ranked among the UniRef50 database. (see Methods 1.6).
	Supplementary Table 7	Query texts are summarized from related literature. Retrieved proteins are mapped to unique IDs. The target protein is ranked among the UniRef50 database. (see Methods 1.6).
	Supplementary Table 8	Query texts are summarized from the UniRef50 database. Retrieved proteins are manually examined with the corresponding webpage in UniProt (keyword match serves as a prerequisite) (see Methods 1.17).

Supplementary Table 3: Matching criteria for these protein-text (protein-to-text and text-to-protein) search tasks.

	Model	Protein-to-text MAP				Text-to-protein MAP			
		Residue-level	Sequence-level	All-level	Global retrieval	Residue-level	Sequence-level	All-level	Global retrieval
Original test set	ProtST	0.059	0.156	0.111	0.007	0.001	0.005	0.003	0.003
	ProteinDT	0.076	0.136	0.108	0.004	0.001	0.001	0.001	0.001
	ProTrek structure encoder	0.274	0.442	0.364	0.216	0.113	0.119	0.116	0.141
	ProTrek sequence encoder	**0.342**	**0.492**	**0.422**	**0.233**	**0.172**	**0.160**	**0.165**	**0.190**
Filtered test set	ProtST	0.043	0.144	0.098	0.009	0.002	0.006	0.004	0.004
	ProteinDT	0.036	0.160	0.104	0.003	0.004	0.002	0.003	0.001
	ProTrek structure encoder	0.275	0.477	0.386	0.203	0.190	0.184	0.186	0.220
	ProTrek sequence encoder	**0.340**	**0.503**	**0.429**	**0.227**	**0.294**	**0.250**	**0.270**	**0.305**

Supplementary Table 4: Results of protein-text retrieval tasks. The "filtered test set" was created by further excluding proteins from both the test set and negative samples (i.e., the 100,000 randomly sampled proteins from UniProt/TrEMBL) that share more than 50% sequence identity with proteins in the 25 million noisy training examples. The number of proteins from Swiss-Prot in test set decreases from 4000 to 2058). The number of negative samples decreases from 100,000 to 63,000.

Protein	Retrieved function	Scoring rank
Om1Cas9 [1]	CRISPR (clustered regularly interspaced short palindromic repeat) is an adaptive immune system that provides protection against mobile genetic elements (viruses, transposable elements and conjugative plasmids). CRISPR clusters contain spacers, sequences complementary to antecedent mobile elements, and target invading nucleic acids. CRISPR clusters are transcribed and processed into CRISPR RNA (crRNA). In type II CRISPR systems correct processing of pre-crRNA requires a trans-encoded small RNA (tracrRNA), endogenous ribonuclease 3 (rnc) and this protein. The tracrRNA serves as a guide for ribonuclease 3-aided processing of pre-crRNA. Subsequently Cas9/crRNA/tracrRNA endonucleolytically cleaves linear or circular dsDNA target complementary to the spacer; Cas9 is inactive in the absence of the 2 guide RNAs (gRNA). Cas9 recognizes the protospacer adjacent motif (PAM) in the CRISPR repeat sequences to help distinguish self versus nonself, as targets within the bacterial CRISPR locus do not have PAMs. PAM recognition is also required for catalytic activity (By similarity). Cuts target DNA in Cas9:gRNAs mixing experiments with C.jejuni strain NCTC 11168 and N.meningitidis strain Z2491.	2
SlugCas9 [2]	Plays a role in repression of expression of endogenous bacterial lipoprotein FTN_1103. Cas9 plays a possibly non-enzymatic role in the degradation of FTN_1103 mRNA, which is dependent on formation of an RNA:RNA complex of tracrRNA and scaRNA (the latter is not found in all type II CRISPR-Cas systems).	1
dsPETase01 [1]	Involved in the degradation and assimilation of the plastic poly(ethylene terephthalate) (PET), which allows I.sakaiensis to use PET as its major energy and carbon source for growth. Likely acts synergistically with PETase to depolymerize PET. Catalyzes the hydrolysis of mono(2-hydroxyethyl) terephthalate (MHET) into its two environmentally benign monomers, terephthalate and ethylene glycol. Does not show activity against PET, bis(hydroxyethyl) terephthalate (BHET), pNP-aliphatic esters or typical aromatic ester compounds catalyzed by the tannase family enzymes, such as ethyl gallate and ethyl ferulate.	1
dsPETase05 [1]		1
dsPETase06 [1]		1
dsPETase02 [1]	Involved in the degradation and assimilation of the plastic poly(ethylene terephthalate) (PET), which allows I.sakaiensis to use PET as its major energy and carbon source for growth. Likely acts synergistically with MHETase to depolymerize PET (PubMed:26965627). Catalyzes the hydrolysis of PET to produce mono(2-hydroxyethyl) terephthalate (MHET) as the major product (PubMed:26965627, PubMed:32269349, PubMed:29666242, PubMed:29603535, PubMed:29374183, PubMed:29235460). Also depolymerizes another semiaromatic polyester, poly(ethylene-2,5-furandicarboxylate) (PEF), which is an emerging, bioderived PET replacement with improved gas barrier properties (PubMed:29666242). In contrast, PETase does not degrade aliphatic polyesters such as polylactic acid (PLA) and polybutylene succinate (PBS) (PubMed:29666242). Is also able to hydrolyze bis(hydroxyethyl) terephthalate (BHET) to yield MHET with no further decomposition, but terephthalate (TPA) can also be observed (PubMed:26965627, PubMed:29603535, PubMed:29374183). Shows esterase activity towards p-nitrophenol-linked aliphatic esters (pNP-aliphatic esters) in vitro (PubMed:26965627, PubMed:30502092)	4
dsPETase03 [1]		2
dsPETase04 [1]		2
TTHA0338: EC 3.13.1.8 [3]	Catalyzes the hydrolysis of S-adenosyl-L-methionine (SAM) into adenosine and L-methionine (PubMed:18551689). Is likely stereoselective, specifically hydrolyzing (R)-S-adenosyl-L-methionine ((R)-SAM), the inactive form of the ubiquitous cofactor SAM, and not the active form of SAM, (S)-S-adenosyl-L-methionine (By similarity). Probaly plays a role in preventing accumulation of (R)-S-adenosyl-L-methionine in cells; maintenance of (S)-S-denosyl-L-methionine homochirality is important for cellular health given that the (R)-form is largely inactive as a methyl donor and can function as an inhibitor of methyltransferases (By similarity). Is unable to mediate a fluorination or chlorination reaction with SAM (PubMed:18551689)	1
SsFlA: EC 2.5.1.94 [3]	Involved in the biosynthesis of the proteosome inhibitor salinosporamide A (SalA). Catalyzes the halogenation of S-adenosyl-L-methionine (SAM) with chloride to generate 5'-chloro-5'-deoxyadenosine (5'-CIDA) and L-methionine. It can also use bromide and iodide, producing halogenated 5'-deoxyadenosine (5'-XDA) and L-methionine, however no halogenase activity is detected in the presence of fluoride	1
$SsdA_{tox}$ [4]	Part of the H2 type VI secretion system (H2-T6SS) specialized secretion system, which delivers several virulence factors in both prokaryotic and eukaryotic cells during infection. Allows the delivery of the phospholipase effector PldA to target cells where it exerts its toxicity	1
$SsdA_{tox}$ - P282S, Y335R, K392E [4]		1
$SsdA_{tox}$ - (260-410) [4]	The TA system protects P.atrosepticum strain 1043 against phage phiM1 and phiA2, E.coli against some but not all coliphages and S.marcescens against some bacteriophages, causing an abortive infection (Abi phenotype) (PubMed:19124776). Also protects P.atrosepticum strain 1043 against phage phiTE; phage that escape Abi and grow in this bacterium have evolved a pseudo-ToxI RNA by expanding a pre-existig sequence similar to the bona fide ToxI repeats (PubMed:23109916)	1
$SsdA_{tox}$ - P282S, Y335R, K392E - (260-410) [4]		1

Supplementary Table 5: Results of protein-to-text retrieval. For each newly validated protein, we utilized ProTrek to predict its natural language functions and recorded the ranking of the first accurately predicted function. For Om1Cas9, the second-ranked retrieved text primarily introduces the fundamental mechanism of the CRISPR-Cas9 system, while the top-ranked text is CRISPR-Cas9 functionally related but not referenced in the literature. Similarly, for dePETases, the retrieved text accurately identifies core functions but also includes descriptions not reported in the literature. We hypothesize that these highly ranked, unreferenced descriptions may indicate potential functions for these proteins, warranting further investigation. Notably, all authentic functions were retrieved within the top five results, with the majority being ranked first. The performance demonstrates ProTrek's effective capability in annotating proteins.

Protein	Protein sequence
Om1Cas9 [1]	UniProtId: A0A2J6P5C0
SlugCas9 [2]	UniProtId: A0A133QCR3
dsPETase01 [1]	Amino Acid Sequence: MKIIRTLGAGIAAVALGFGLLAPSASVAAENDYERGPDPTSSSIEASRGPYAVSTKSISRFAARGF GGGTIHYPTTTADGTFGVVAVSPGYTASESTIRWLGPRLASFGFVVITFDTNSRYDQPRARGTQ LLAAIDQAIGDSTVGSRIDPSRQAVVGHSMGGGGTLEAAKTRPSIEAAVGLTPWNLDKTWPEV EAAALQIGAQNDSVAPPRSHAVPFYGSLTNAERRAYLELRGASHFAPNTSNTTIAKYTLAWLKR YVDDDTRYEQFLAPGPSTGFGSAVSDYRIQ
dsPETase02 [1]	Amino Acid Sequence: MNLLRIKTLSSALIGLVILAGSFTTPSTVAQDNSDGRYRPEGRTFARNAANMFDDRVFEARTYE TGTNQEAFASATIFYPLTLSFDPPNGAVIMVPGYRGTPPVYDWWGPMLASIGVITMIIETNTPE DSLEARKNAFIAGVEFLRGENNNADSPIRDKLDTGNIAIMGHSLGGGASLRAAEELASQIKAVI PLTPYCCELGQPFEGDLSGVSVPTLIIASAEDAIAPPGDHARMLYDSVNASTKKVYLEFATGNH MLPSNSGQDLETLGTYVYAFIKENFTDNPRYTDFIFGDGEEQFSIYETNQ
dsPETase03 [1]	Amino Acid Sequence: MLLALLFAACSTSVTNRTGDPNISIASLSADGPYAVRAYTSFPEVGEFADATIYYPLDAETPV GGVAVSPGFTELQRAINWWGPRLASHGFAVLILDTNEPRDSPELRAEALISAVRLLKAENSRP DSPLNGRIDVNKMAIMGHSMGGGGTLIAANNYSDEIQAAIPFTPWQPEGDFSQITVPTLVIAG AADRIAAVSDHAWPHYQSLPHSTTRVYLEVAGGNHFLANSSGPDLSTIGRYGIAWLKLYLDGD ERYRDFIYGEAQKVDTGKFSRYIANP
dsPETase04 [1]	Amino Acid Sequence: MANPYERGPNPTDALLEASSGPFSVSEENVSRLSASGFGGGTIYYPRENNTYGAVAISPGYT MNRWLFPRALLLLFSLLLASCANSPQPTVESLSGDGEYQVMTYTDFPDVPEFGDATIYYPLD TRGSIGGVAISPGYTERQSHIEWWGPLLASHGYAVLVLDTNDRRESTDLRADALIAAVTTLR AENTRNDSPLMGRIDGGKMAIMGHSMGGGGTLIAAHEHGEEIQAAIPFTPWEPDGPFDNITV PTLIIAGSIDRIAGVDEHAWRHFQSIPESTTKVYMEIDGGNHYIADTDRGTDLATVGRYGIA WLKLYLDGDERYRDFIYGEYHTPDMEKFSRYVTNP
dsPETase05 [1]	Amino Acid Sequence: MKSGQHQKKDTVMKTPLFKLAALTLGVSLSSVALATNPGGGGGGSNPDTGTGFPGVSSFSA DGSFATTSGSAGLSCTVFRPSTLGANGLKHPIIVWGNGTTASPSTYSGILEHWASHGFVVI AANTSNAGTGQDMLNCVDYLTTQNNRSTGTYANKLDLNRIGAAGHSQGGGGTIMAGQDYRI KVTAPFQPYTIGLGHNSSSQSNQNGPMFLMTGSADTIASPTLNALPVYNRANVPVFWGELS GASHFEPVGSAGDFRGPSTAWFRYHLMDDASAEDTFYGSNCDLCTDNDWDVRRKGIN
dsPETase06 [1]	Amino Acid Sequence: MSVSLKGALRVLPLAASVALVGCFGGGDPEPGPDGQALTNPGEYEICSYETDLENSGYAS ARMTYPCDLSDGPYPATTLTGGFTNTKEQMEWLAEHLTTHGYVVLTMTPNNTLGVPPGWR DAQLGGFAELADENARSNSPLKGKIDLSKRNIMGFSMGGGGVILAAEEMGDAPTSAIALA PWLGAYNVDYSQIETPMLMLGSENDELAYYTEDYYAQLPADLERGVAIYAGASHFDWYGV NNQDQKAQFRTLVTAFLEVQLKGDTSAYSYFDGAEHDEHVQEGWFSAFDYQK
TTHA0338 [3]	UniProtId: Q5SLF5
SsFIA [3]	UniProtId: W0W999
$SsdA_{tox}$ [4]	UniProtId: A0A3M2XVC5

Supplementary Table 6: Protein sequences extracted from recent literature for validating ProTrek's text-protein translation capabilities.

Category	Text description	Protein	Rank/4.5×10^7
CRISPR-Cas9	The protein (CRISPR-Cas9 system) utilises a guide RNA scaffold consisting of a 37-bp mature CRISPR RNA (crRNA) and a 72-bp trans-activating crRNA (tracrRNA) and specifically recognizes 3' NNGG protospacer adjacent motif (PAM) sequences for targeting of double-stranded DNA (dsDNA).	Om1Cas9 [1]	165
Compact CRISPR-Cas9	Compact Cas9 refers to a smaller version of the Cas9 enzyme, often derived from bacteria with smaller genomes like Staphylococcus aureus (SaCas9) or other similar organisms. Compared to the widely used Streptococcus pyogenes Cas9 (SpCas9), which is relatively large (4.2 kb), compact Cas9 variants are much smaller, typically around 3 kb or less. This smaller size makes compact Cas9s more suitable for delivery in gene therapy applications, especially when using viral vectors like adeno-associated viruses (AAVs), which have strict packaging limits. Despite being smaller, compact Cas9 enzymes retain the ability to make targeted double-stranded breaks in DNA and can be engineered or adapted to function similarly to SpCas9 in various CRISPR applications, such as genome editing, gene regulation, and base editing.	SlugCas9 [2]	24
PETase	This protein is identified as PET hydrolase.	dsPETase01 [1] dsPETase02 [1] dsPETase03 [1] dsPETase04 [1] dsPETase05 [1] dsPETase06 [1]	24 167 659 271 7 272
EC 3.13.1.8	The enzyme, (R)-S-adenosyl-L-methionine hydrolase (EC 3.13.2.3), is classified as a hydrolase, specifically acting on carbon-sulfur bonds. It catalyzes the conversion of (R)-S-adenosyl-L-methionine and water into adenosine and L-methionine. This reaction is essential in some bacteria and archaea for removing the (R) isomer of S-adenosyl-L-methionine from the cell through a nucleophilic attack of water at the C5' carbon of S-adenosyl-L-methionine, resulting in the formation of adenosine and L-methionine.	TTHA0338 [3] SsFlA [3]	194 433
EC 2.5.1.94	The enzyme, Adenosyl-chloride synthase (EC 2.5.1.94), is a transferase that facilitates the reaction of S-adenosyl-L-methionine with chloride to produce 5'-deoxy-5'-chloroadenosine and L-methionine. It is categorized under transferases that transfer alkyl or aryl groups, other than methyl groups. This enzyme is key in the biosynthesis of the proteasome inhibitor salinosporamide A and is isolated from the marine bacterium Salinispora tropica. It is closely related to adenosyl-fluoride synthase (EC 2.5.1.63) but does not accept fluoride as a substrate.	SsFlA [3]	3
EC 2.5.1.63	The enzyme, Adenosyl-fluoride synthase (EC 2.5.1.63), is classified under transferases, specifically those transferring alkyl or aryl groups other than methyl groups. It catalyzes the reaction between S-adenosyl-L-methionine and fluoride to form 5'-deoxy-5'-fluoroadenosine and L-methionine. This type of reaction is critical in certain biological processes in bacteria and archaea, involving the transfer of a fluorine atom to an adenosine molecule.	SsFlA [3]	74
Deaminase	A bacterial toxin derived from Pseudomonas syringae that enables targeted cytosine base editing.	$SsdA_{tox}$ [4] $SsdA_{tox}$ - (260-410) [4] $SsdA_{tox}$ - P282S,Y335R,K392E [4] $SsdA_{tox}$ - P282S, Y335R, K392E - (260-410) [4]	422 106 533 149

Supplementary Table 7: Results of text-to-protein retrieval. To simulate the search process, we added these newly characterized proteins to the UniRef50 database, which contains a total of 45 million protein sequences. Subsequently, we searched for relevant proteins based on the textual descriptions summarized from their papers. For $SsdA_{tox}$, the DYW-like deaminase toxin domain (the last 151 amino acids of $SsdA_{tox}$)—representing the main functional segment (sytosine base editing)—is ranked higher than the full-length $SsdA_{tox}$ sequence based on the description of $SsdA_{tox}$. This emphasizes ProTrek's capability to identify localized functional sequences with high confidence based on specific functional descriptions. Notably, these newly validated proteins ranked very high in their respective query texts, highlighting ProTrek's promising potential as a text-oriented protein retrieval system.

Protein category	Fine-grained subclass description	Top 20 match count
CRISPR	**Large CRISPR** Large CRISPR systems, exemplified by the traditional CRISPR-Cas9 system, typically involve larger proteins, such as Cas9, that require more space for delivery. These systems are highly versatile and capable of cutting DNA at specific sites, enabling a wide array of genome-editing tasks, from gene insertions to larger-scale modifications. Due to their size, they often require more sophisticated delivery methods, especially in therapeutic applications. Large CRISPR systems are well-established for research in gene therapy, functional genomics, and synthetic biology because of their robust cutting and editing abilities and the wealth of available tools and protocols for their application.	20 / 20
	Small CRISPR Small CRISPR systems, often referred to as compact or mini-CRISPR systems, are more streamlined versions of CRISPR-Cas technology, featuring smaller proteins like Cas12, Cas13, and Cas14. These systems are particularly advantageous for applications requiring delivery into cells or tissues with size constraints, such as using viral vectors (e.g., AAV) that have limited packaging capacities. Despite their smaller size, they maintain the ability to perform precise genome editing and, in the case of Cas13, even target RNA for editing. Cas14, being one of the smallest CRISPR systems, is especially compact and suitable for applications where space is at a premium. These small systems are often optimized for single-nucleotide changes or gene knockdowns and tend to have a high specificity for target DNA or RNA sequences, reducing off-target effects in many cases. Their small size and versatility make them promising tools in gene therapy, functional genomics, and synthetic biology.	20 / 20
Deaminase	**Adenosine deaminase** Adenosine deaminase (ADA) is an enzyme that plays a crucial role in purine metabolism by catalyzing the deamination of adenosine, converting it into inosine. This reaction is essential for maintaining proper cellular function and balancing nucleotide pools. By facilitating the conversion of adenine to inosine, adenosine deaminase contributes to cellular processes that involve nucleotide synthesis and recycling, which are important for DNA and RNA stability and function.	20 / 20
	Cytosine deaminase Cytosine deaminase is an enzyme that catalyzes the conversion of cytosine to uracil by removing an amine group. This reaction is significant for nucleotide metabolism and can play roles in cellular regulation, including RNA processing and the maintenance of nucleotide balance. Cytosine deaminase activity is involved in biochemical pathways that impact gene expression and the metabolic processing of cytosine-containing compounds.	18 / 20
Family	**APOBEC** The APOBEC family comprises cytidine deaminases that catalyze the conversion of cytidine (C) to uracil (U) in RNA or DNA sequences. APOBEC proteins play critical roles in processes such as RNA editing, viral defense, and immune response. In genome editing, APOBEC1 is often engineered into Cytosine Base Editors (CBEs) to precisely convert C-to-T in DNA without causing double-strand breaks. These editors are widely used to correct point mutations, study gene function, and explore therapeutic interventions, particularly in applications targeting C•G to T•A base pair conversions.	20 / 20
	TET The TET (Ten-Eleven Translocation) family consists of enzymes that mediate DNA demethylation by oxidizing 5-methylcytosine (5mC) to 5-hydroxymethylcytosine (5hmC) and further oxidation derivatives. This family is essential for epigenetic regulation, impacting processes such as cellular differentiation, gene expression, and development. TET proteins do not perform direct base editing but influence gene activity through epigenetic modifications. They are of particular interest in cancer research, where abnormal DNA methylation patterns play a key role in tumor development and progression.	20 / 20
	ADAR The ADAR (Adenosine Deaminases Acting on RNA) family is responsible for converting adenosine (A) to inosine (I) in double-stranded RNA. Inosine is interpreted as guanosine (G) during translation, altering the RNA's coding potential and thereby influencing gene expression. ADAR proteins are particularly crucial in neural development and immune system regulation. While ADARs target RNA instead of DNA, their A-to-I RNA editing mechanism makes them valuable tools for studying gene regulation and for potential therapeutic applications, especially in diseases caused by RNA mutations.	11 / 20
	FTO/ALKBH The FTO (Fat mass and obesity-associated) and ALKBH (AlkB homolog) family of proteins are involved in the regulation of RNA and DNA methylation, particularly by demethylating N6-methyladenosine (m6A) in RNA and other methylated bases. FTO and ALKBH proteins are critical in modulating mRNA stability, translation, and splicing, making them key players in post-transcriptional gene regulation. These demethylases are linked to various biological processes such as obesity, cancer, and neurodevelopment. Their ability to reverse methyl modifications on nucleic acids provides valuable insights into the dynamic regulation of gene expression and offers therapeutic potential in diseases driven by epigenetic dysregulation.	20 / 20
	DNMT The DNMT (DNA Methyltransferase) family is responsible for adding methyl groups to cytosine residues in DNA, generating 5-methylcytosine (5mC), a key epigenetic mark involved in gene silencing. DNMT1, DNMT3A, and DNMT3B are the main enzymes in this family, playing critical roles in maintaining and establishing DNA methylation patterns during development and cell division. This family is crucial for regulating gene expression, cellular differentiation, and genomic imprinting. DNMT dysfunction is associated with various cancers, where abnormal methylation patterns contribute to tumor progression. Due to their role in epigenetic regulation, DNMTs are important targets in cancer therapy and the study of gene expression control.	19 / 20

Protein category	Textual description of subcategories	Top 20 match count
	UNG Uracil-DNA Glycosylases (UNGs) are enzymes involved in DNA repair, specifically in the base excision repair (BER) pathway. They recognize and remove uracil residues from DNA, which can arise through deamination of cytosine or through incorporation of dUMP during DNA replication. By cleaving the glycosidic bond between the uracil and the sugar-phosphate backbone, UNG initiates a cascade of repair processes that correct the error. This family plays a critical role in maintaining genomic stability by preventing mutagenesis caused by uracil misincorporation. While UNG enzymes are not used directly for genome editing, they are essential for ensuring the accuracy of genetic information during replication and repair.	20 / 20
	FEN1 Flap Endonuclease 1 (FEN1) is an essential enzyme in the process of DNA replication and repair, particularly in the resolution of Okazaki fragments during lagging strand synthesis. FEN1 recognizes and cleaves the "flap" structures formed by displaced single-stranded DNA during replication. It is also involved in long-patch base excision repair (BER), where it removes displaced DNA strands during the repair of damaged bases. FEN1 ensures the integrity of the DNA replication process, preventing mutations and maintaining genome stability. FEN1's precise DNA cleavage function is critical for DNA metabolism, but it has also been investigated for its potential in biotechnological applications involving DNA manipulation.	19 / 20
	SMUG SMUG1 is a glycosylase enzyme that plays a role in the base excision repair (BER) pathway by removing uracil from single-stranded DNA (ssDNA). Unlike UNG, which primarily removes uracil from double-stranded DNA, SMUG1 specializes in targeting uracil in single-stranded regions, such as those formed during transcription or replication. SMUG1 is also involved in the removal of other damaged or mismatched bases, including oxidized pyrimidines. Its activity ensures that errors introduced during DNA replication and transcription are efficiently corrected, contributing to genome stability and preventing mutagenesis. SMUG1 has been studied for its role in both normal cellular processes and in cancer, where DNA repair pathways are often disrupted.	20 / 20
Organism	**Bacteria** Bacteria are a vast group of single-celled prokaryotic organisms that lack a membrane-bound nucleus. They are found in virtually every environment on Earth, from soil to oceans, and even within other organisms. Bacteria play critical roles in various ecological processes, such as nitrogen fixation, decomposition, and symbiotic relationships with plants and animals. They are also the basis for many biotechnological applications, including antibiotic production, fermentation, and genetic engineering. Some bacteria are pathogenic, while others are essential for maintaining life through processes like digestion and nutrient cycling.	20 / 20
	Archaea Archaea are single-celled prokaryotic organisms similar to bacteria but with distinct evolutionary and biochemical differences. They are known for thriving in extreme environments such as hot springs, salt lakes, and deep-sea hydrothermal vents. Archaea have unique cell membranes and metabolic pathways that allow them to survive in harsh conditions, such as high temperatures, acidity, and salinity. Despite their extremophilic nature, they also inhabit more moderate environments and play important roles in biogeochemical cycles, such as methane production and nitrogen cycling.	19 / 20
	Eukaryotes Eukaryotes are organisms with complex cells that contain a true nucleus enclosed by a membrane, as well as various membrane-bound organelles such as mitochondria and chloroplasts. Eukaryotes include all plants, animals, fungi, and protists. This domain is characterized by its diversity and complexity in form and function, with organisms ranging from single-celled amoebas to multicellular organisms like humans. Eukaryotes engage in more specialized biological processes due to their compartmentalized cell structure, such as oxidative phosphorylation, photosynthesis, and sophisticated developmental systems.	20 / 20
CRISPR + Organism	**Cas1 + Archaea** Cas1 is a key protein in the CRISPR-Cas system of Archaea, playing a crucial role in the adaptation phase of the immune response. It integrates viral DNA fragments into the CRISPR array, allowing the organism to build a genetic memory of past viral infections. This memory is then used to recognize and destroy the same viruses during future encounters. In archaea, Cas1 helps maintain genome stability and provides an essential defense mechanism, especially in extreme environments where these organisms often thrive.	11 / 20
	Cas1 + Bacteria Cas1 is a central component of the CRISPR-Cas system in Bacteria, responsible for integrating viral DNA fragments into the CRISPR array during the adaptation phase. This process allows bacteria to "remember" past viral infections and mount a targeted defense in future attacks. Cas1 ensures that the CRISPR system remains functional, providing an adaptive immune response that helps protect against invading genetic elements such as bacteriophages.	20 / 20
	Cas3 + Archaea In Archaea, the Type I CRISPR-Cas system, particularly involving Cas3, is a key defense mechanism against invading genetic elements like viruses. Cas3 is an enzyme with both helicase and nuclease activities, capable of degrading foreign DNA in a processive manner. In archaeal systems, the Cas3 protein works with a multi-subunit Cascade complex to recognize and unwind target DNA, providing a robust adaptive immune response in extreme environments where archaea typically thrive.	11 / 20

Protein category	Textual description of subcategories	Top 20 match count
	Cas3 + Bacteria In Bacteria, the Type I CRISPR-Cas system is the most widespread CRISPR variant, with Cas3 playing a central role in defending against phages and plasmids. Cas3 functions by degrading foreign DNA after it has been identified by the Cascade protein complex. Its ability to progressively degrade DNA makes it an effective defense system in bacterial cells, enabling them to protect against and eliminate genetic invaders in a highly efficient manner.	20 / 20
	Cmr/Csm Complex + Archara In Archaea, the Type III CRISPR-Cas system, with the Cmr/Csm complex, targets foreign RNA and, in some cases, degrades associated DNA. This dual-targeting capability is essential for defending against diverse viral threats, especially in the extreme environments where archaea thrive. The system's ability to act on both RNA and DNA provides a robust and adaptive immune defense.	7 / 20
	Cmr/Csm Complex + Bacteria In Bacteria, the Type III CRISPR-Cas system, featuring the Cmr/Csm complex, plays a critical role in immune defense by targeting invading RNA and sometimes degrading linked DNA. This system helps bacterial cells neutralize viral RNA, preventing replication and ensuring effective protection against a wide range of genetic invaders. Its versatility makes it a vital component of bacterial adaptive immunity.	14 / 20
Deaminase + Organism	**Adenosine deaminase + Archaea** In archaea, adenosine deaminase (ADA) is an enzyme that plays a crucial role in purine metabolism by catalyzing the deamination of adenosine, converting it into inosine. This reaction is essential for maintaining proper cellular function and balancing nucleotide pools. By facilitating the conversion of adenine to inosine, adenosine deaminase in archaea contributes to cellular processes that involve nucleotide synthesis and recycling, which are vital for the stability and function of archaeal DNA and RNA.	10 / 20
	Adenosine deaminase + Bacteria In bacteria, adenosine deaminase (ADA) functions as an important enzyme in purine metabolism, catalyzing the deamination of adenosine to produce inosine. This enzymatic reaction is critical for balancing nucleotide pools and ensuring proper cellular function. By converting adenine to inosine, adenosine deaminase supports bacterial cellular processes involving nucleotide synthesis and recycling, which are important for DNA and RNA integrity and stability.	17 / 20
	Adenosine deaminase + Eukaryota In eukaryota, adenosine deaminase (ADA) is an essential enzyme involved in purine metabolism, catalyzing the deamination of adenosine to convert it into inosine. This reaction plays a critical role in maintaining nucleotide pool balance and supporting proper cellular function. By facilitating the conversion of adenine to inosine, adenosine deaminase contributes to nucleotide synthesis and recycling, processes crucial for the stability and function of eukaryotic DNA and RNA. Mutations or deficiencies in ADA activity in eukaryotes, such as humans, can lead to severe metabolic disorders like severe combined immunodeficiency (SCID).	18 / 20
	Cytosine deaminase + Archaea In archaea, cytosine deaminase is a crucial enzyme that catalyzes the conversion of cytosine to uracil by removing an amine group. This enzymatic activity is particularly significant in the unique metabolic pathways of archaeal organisms, contributing to nucleotide metabolism and various cellular regulatory functions. It plays a vital role in RNA processing, maintaining nucleotide balance, and facilitating the efficient adaptation of archaea to extreme environments. The presence and activity of cytosine deaminase in archaea directly impact gene expression and support the metabolic processing of cytosine-containing compounds, which are essential for maintaining the stability and functionality of archaeal cellular processes.	8 / 20
	Cytosine deaminase + Bacteria In bacteria, cytosine deaminase is an enzyme responsible for catalyzing the conversion of cytosine to uracil through the removal of an amine group. This reaction is essential for nucleotide metabolism and contributes to the regulation of cellular processes, such as RNA editing and maintaining nucleotide balance. Bacterial cytosine deaminase plays a role in pathways that affect gene expression and the processing of cytosine-containing compounds, supporting bacterial growth and survival.	8 / 20
	Cytosine deaminase + Eukaryota In eukaryota, cytosine deaminase is an enzyme that catalyzes the deamination of cytosine to uracil by removing an amine group. This enzymatic process is important for nucleotide metabolism and cellular regulation, influencing RNA processing and nucleotide balance. Eukaryotic cytosine deaminase activity is involved in pathways that affect gene expression and the metabolism of cytosine-containing compounds, contributing to cellular functions such as mRNA editing and the synthesis of nucleic acid components.	14 / 20

Supplementary Table 8: Result of fine-grained text-to-protein retrieval. Textual descriptions of subcategories were used to query the UniProt50 database, retrieving the top 20 proteins, all excluded from the training set to ensure unbiased evaluation. The number of proteins matching the specified fine-grained textual descriptions was then assessed for accuracy.

EC Number	Textual description of EC number
EC 1.4.3.4	EC 1.4.3.4: The enzyme, Monoamine oxidase, is an oxidoreductase that catalyzes the oxidative deamination of monoamines, which are neurotransmitters and biogenic amines. It utilizes oxygen as an electron acceptor to convert primary, secondary, and some tertiary amines into aldehydes, ammonia, and hydrogen peroxide. The enzyme is a flavoprotein found in the outer mitochondrial membrane and plays a significant role in the metabolism of neurotransmitters such as serotonin, dopamine, and norepinephrine. It differs from other amine oxidases by its ability to oxidize secondary and tertiary amines. Monoamine oxidase is inhibited by specific compounds like clorgyline, 1-deprenyl, and pargyline, but is not inhibited by semicarbazide, which makes it distinct from similar enzymes like diamine oxidase (EC 1.4.3.22) and primary-amine oxidase (EC 1.4.3.21).
EC 2.7.1.1	EC 2.7.1.1: The enzyme, Hexokinase, is a transferase that catalyzes the phosphorylation of D-hexoses using ATP as a phosphate donor, resulting in the production of D-hexose 6-phosphate and ADP. It belongs to the group of phosphotransferases with an alcohol group as the acceptor. This enzyme can act on various substrates, including D-glucose, D-mannose, D-fructose, sorbitol, and D-glucosamine, with ITP and dATP as alternative phosphate donors. The liver isoenzyme, often referred to as glucokinase, is distinct for its role in glucose metabolism. Hexokinase plays a central role in glycolysis by activating sugars for subsequent metabolic pathways.
EC 3.1.3.1	EC 3.1.3.1: The enzyme, Alkaline phosphatase, is a hydrolase that catalyzes the hydrolysis of phosphate monoesters into an alcohol and phosphate under alkaline conditions. It exhibits wide substrate specificity and can also catalyze transphosphorylations. The human placental version of this enzyme is a zinc protein. Some versions of the enzyme are capable of hydrolyzing diphosphates, distinguishing it from similar enzymes like EC 3.6.1.1 (inorganic diphosphatase).
EC 4.1.1.39	EC 4.1.1.39: The enzyme, ribulose-bisphosphate carboxylase/oxygenase (RuBisCO), is a carboxylyase responsible for catalyzing the conversion of 3-phospho-D-glycerate and H+ into D-ribulose 1,5-bisphosphate, CO2, and H2O. It plays a central role in the Calvin-Benson cycle of photosynthesis by fixing atmospheric CO2. RuBisCO can also utilize O2 instead of CO2, resulting in the formation of 3-phospho-D-glycerate and 2-phosphoglycolate, a process linked to photorespiration. The enzyme's activity and specificity for CO2 versus O2 are critical for the efficiency of carbon fixation in plants and other autotrophic organisms.
EC 5.3.1.9	EC 5.3.1.9: The enzyme, glucose-6-phosphate isomerase, is an isomerase that catalyzes the reversible conversion of alpha-D-glucose 6-phosphate into beta-D-fructofuranose 6-phosphate. It is classified as an intramolecular oxidoreductase, interconverting aldoses and ketoses. In addition to this reaction, the enzyme can catalyze the anomerization of D-hexose 6-phosphates. Found in yeast, it plays a significant role in sugar metabolism by ensuring the interconversion of these essential sugar phosphates in metabolic pathways.
EC 6.1.1.1	EC 6.1.1.1: The enzyme, tyrosine-tRNA ligase, is a ligase that forms carbon-oxygen bonds by catalyzing the aminoacylation of tRNATyr with L-tyrosine. It utilizes ATP to form L-tyrosyl-tRNATyr and produces AMP and diphosphate as byproducts. This enzyme belongs to the class of ligases forming aminoacyl-tRNA and related compounds. The systematic name of the enzyme is L-tyrosine:tRNATyr ligase (AMP-forming). It plays a crucial role in protein biosynthesis by ensuring that L-tyrosine is accurately attached to its corresponding tRNA molecule, tRNATyr. The reaction it catalyzes is ATP + L-tyrosine + tRNATyr = AMP + diphosphate + L-tyrosyl-tRNATyr.

Supplementary Table 9: Selected examples from standardized textual description database of EC numbers. The prompt used for generating EC descriptions is as follows: "Summarize the following text using the specified template format. Ensure the output is formatted exactly as shown in the template, and do not include history information. Enclose the final output in square brackets. Template format: *template output*. Text to summarize: *text*." Here, *template output* denotes one of the example standardized description shown in this table, and *text* corresponds to raw EC information to be summarized.

Task	**Metric**	ESM-2	ProtST-ESM-1b	ProtST-ESM-2	ESM-3	**ProTrek**
Thermostability	Spearman's ρ	0.673	0.692	0.672	0.699	0.697
Subcellular Localization	Acc%	84.67	83.92	85.60	81.29	88.52
Binary Localization	Acc%	93.56	93.80	94.42	94.05	95.42
Structure Class	Acc%	79.90	81.09	80.80	81.59	82.19
Structural Similarity	Spearman's ρ	0.677	0.671	0.679	0.678	0.677
Binding Site Detection	Mcc	0.540	0.501	0.554	0.558	0.536
Metal Ion Binding	Acc%	74.97	79.03	77.61	78.72	79.58
Fluorescence	Spearman's ρ	0.698	0.696	0.701	0.700	0.702
Stability	Spearman's ρ	0.780	0.751	0.755	0.778	0.760
β-lactamase	Spearman's ρ	0.904	0.897	0.908	0.917	0.912
AAV	Spearman's ρ	0.813	0.803	0.829	0.839	0.834

Supplementary Table 10: Comprehensive benchmark results for downstream fine-tuning tasks. Spearman's ρ, Acc%,, and MCC represent Spearman's rank correlation coefficient, classification accuracy, and Matthews Correlation Coefficient, respectively.

EC Number	Textual description of EC number
EC 1.4.3.4	The enzyme, Monoamine oxidase (EC 1.4.3.4), is an oxidoreductase that catalyzes the oxidative deamination of monoamines, which are neurotransmitters and biogenic amines. It utilizes oxygen as an electron acceptor to convert primary, secondary, and some tertiary amines into aldehydes, ammonia, and hydrogen peroxide. The enzyme is a flavoprotein found in the outer mitochondrial membrane and plays a significant role in the metabolism of neurotransmitters such as serotonin, dopamine, and norepinephrine. It differs from other amine oxidases by its ability to oxidize secondary and tertiary amines. Monoamine oxidase is inhibited by specific compounds like clorgyline and pargyline, which makes it distinct from similar enzymes like diamine oxidase.
EC 2.7.1.1	The enzyme, Hexokinase (EC 2.7.1.1), is a transferase that catalyzes the phosphorylation of D-hexose (such as glucose) using ATP, converting it into D-hexose 6-phosphate. This reaction is crucial for glycolysis, as it traps glucose within the cell and begins its metabolic breakdown. Hexokinase acts on various hexoses like D-glucose, D-mannose, and D-fructose, and is involved in key metabolic pathways, including glycolysis, gluconeogenesis, and fructose metabolism. Some isoenzymes, like glucokinase, are liver-specific and have specialized roles in glucose regulation. This enzyme plays an essential role in energy metabolism and sugar processing.
EC 3.1.3.1	The enzyme, Alkaline phosphatase (EC 3.1.3.1), is a hydrolase that catalyzes the hydrolysis of phosphate monoesters, producing an alcohol and inorganic phosphate. This enzyme has wide substrate specificity and can also catalyze transphosphorylation reactions. Alkaline phosphatase is a zinc-dependent enzyme, and its human placental form is a prominent example. It plays key roles in phosphate metabolism and is involved in pathways such as thiamine metabolism and folate biosynthesis. This enzyme is found in many tissues and is important for processes like bone mineralization and digestion.
EC 4.1.1.39	The enzyme, Ribulose-bisphosphate carboxylase (EC 4.1.1.39), commonly known as RuBisCO, is a carboxy-lyase that catalyzes the carboxylation of D-ribulose 1,5-bisphosphate, a key step in the Calvin cycle of photosynthesis. It converts 3-phospho-D-glycerate into D-ribulose 1,5-bisphosphate with the release of carbon dioxide (CO2) and water (H2O). RuBisCO can also catalyze an oxygenation reaction when O2 is used instead of CO2, leading to the formation of 2-phosphoglycolate. This enzyme is essential for carbon fixation in plants, algae, and certain microorganisms. RuBisCO is a critical enzyme in the global carbon cycle and is involved in pathways such as carbon fixation and glyoxylate metabolism.
EC 5.3.1.9	The enzyme, Glucose-6-phosphate isomerase (EC 5.3.1.9), is an isomerase that catalyzes the reversible conversion between alpha-D-glucose 6-phosphate and beta-D-fructofuranose 6-phosphate. This enzyme plays a critical role in glycolysis and gluconeogenesis by facilitating the interconversion of glucose and fructose phosphates, which are essential for energy metabolism. It is involved in several metabolic pathways, including the pentose phosphate pathway, starch and sucrose metabolism, and amino sugar metabolism. Glucose-6-phosphate isomerase is found in a wide range of organisms, including archaea, bacteria, and eukaryotes, and it contributes to the efficient use of carbohydrates in cellular metabolism.
EC 6.1.1.1	The enzyme, Tyrosine-tRNA ligase (EC 6.1.1.1), is a ligase that catalyzes the reaction between ATP, L-tyrosine, and tRNA(Tyr) to form L-tyrosyl-tRNA(Tyr), AMP, and diphosphate. This enzyme plays a critical role in protein synthesis by linking tyrosine to its corresponding tRNA, enabling its incorporation into growing polypeptide chains. It is classified under ligases that form carbon-oxygen bonds, specifically aminoacyl-tRNA ligases. Tyrosine-tRNA ligase is involved in the aminoacyl-tRNA biosynthesis pathway, essential for accurate translation of the genetic code.

Supplementary Table 11: Textual descriptions of specific EC numbers for text-to-sequence retrieval.

CRISPR-associated protein	Textual description of CRISPR-associated proteins
CRISPR-Cas3	Cas3 is part of the Type I CRISPR-Cas systems, serving as a signature protein with both nuclease and helicase activities that play a critical role in degrading foreign DNA. It is integral to the immune response against invading genetic material but is less commonly applied in gene-editing applications due to its complex mechanism. Its dual function enables it to unwind and cleave DNA, contributing to the degradation pathway essential for microbial defense.
CRISPR-Cas9	Cas9 is part of the Type II CRISPR-Cas systems, functioning as a single-protein system that specifically targets DNA. It was initially discovered in Streptococcus pyogenes and is renowned for its ability to make precise cuts in DNA at target sites defined by guide RNA. Cas9 generates blunt ends when it cuts DNA, making it highly suitable for creating gene knockouts or insertions at specific locations. The ease of designing guide RNAs for Cas9 allows for targeted genome editing with high specificity, enabling applications across a wide range of organisms. Its reliability and precision have made Cas9 one of the most widely used tools in genetic engineering, biotechnology, and therapeutic research.
CRISPR-Cas10	Cas10 is part of the Type III CRISPR-Cas systems, serving as the signature protein that facilitates complex antiviral defense mechanisms. It is a multi-domain protein that plays a crucial role in targeting both RNA and, indirectly, DNA. Cas10 is unique in its dual activity, where it binds CRISPR RNA (crRNA) to guide the recognition of target RNA and induces transcription-coupled DNA cleavage through an auxiliary complex. This capability allows Cas10 to participate in more intricate gene regulation processes, including immune response signaling and the modulation of foreign genetic material. The multifunctionality of Cas10 has made it an intriguing subject of study, especially for applications that require a combined approach to RNA and DNA interference. Its catalytic versatility extends the potential use of Type III CRISPR systems in advanced genetic engineering and synthetic biology research.
CRISPR-Cas12	Cas12 is part of the Type V CRISPR-Cas systems, functioning as a single-protein system that specifically targets DNA. It was initially discovered in organisms such as Acidaminococcus and Lachnospiraceae. Cas12 generates staggered (sticky) ends when it cuts DNA, which facilitates the insertion of specific sequences and can be beneficial for DNA repair. An additional unique feature of Cas12 is its ability to independently process CRISPR RNA (crRNA) from a precursor transcript. This capability enables Cas12 to perform multiplex gene editing efficiently, making it especially useful in complex genetic engineering applications that require editing multiple genes at once. Its distinctive cutting mechanism and versatility in DNA editing have made Cas12 a powerful tool in biotechnological and therapeutic fields.
CRISPR-Cas13	Cas13 is a distinctive member of the Type VI CRISPR-Cas systems, notable for its ability to exclusively target and cleave RNA rather than DNA. This protein operates by recognizing specific RNA sequences using guide RNA (crRNA) and initiating precise RNA cleavage, accompanied by collateral RNA degradation. Cas13's unique RNA-targeting mechanism makes it a powerful tool for RNA editing, gene expression regulation, and antiviral applications.
CRISPR Csm/Cmr Complex	The Csm/Cmr complex is part of the Type III CRISPR-Cas systems, functioning as a multi-protein system that can target both DNA and RNA. It is commonly found in various bacteria and archaea, including thermophiles. The Csm/Cmr complex operates by recognizing and degrading foreign genetic material from viruses or plasmids, contributing to cellular immune defense. Its unique ability to target both types of nucleic acids allows it to defend against a wider range of threats. This dual functionality makes the Csm/Cmr complex an important model for understanding immune defense mechanisms and offers potential applications in biotechnological fields that require simultaneous RNA and DNA targeting.

Supplementary Table 12: Textual descriptions of CRISPR-associated proteins for text-to-sequence retrieval.

Model	Metric	TM-score								
		0.1-0.2	0.2-0.3	0.3-0.4	0.4-0.5	0.5-0.6	0.6-0.7	0.7-0.8	0.8-0.9	0.9-1.0
BLASTP	TP	1717	2738	2131	1322	2022	10220	6751	57479	222120
	FP	584	4550	5245	1339	177	170	52	40	34
	Precision	0.746	0.376	0.289	0.497	0.920	0.984	0.992	0.999	0.999
DIAMOND	TP	16	111	298	270	1618	9942	5928	55594	220988
	FP	0	0	14	4	5	4	4	14	25
	Precision	1.000	1.000	0.955	0.985	0.997	0.999	0.999	0.999	0.999
MMseqs2	TP	61	385	720	576	1775	9960	6184	55566	221873
	FP	0	4	2	4	5	22	10	29	33
	Precision	1.000	0.990	0.997	0.993	0.997	0.998	0.998	0.999	0.999
Foldseek	TP	3811	5827	5259	3170	4096	11104	7190	57722	222092
	FP	394	3631	6843	4633	1331	1676	559	69	33
	Precision	0.906	0.616	0.435	0.406	0.755	0.869	0.928	0.999	0.999
ProTrek structure-to-structure	TP	20645	70185	63827	16592	8057	10100	6321	54781	218837
	FP	997	7394	8814	1556	474	157	60	21	27
	Precision	0.954	0.905	0.879	0.914	0.944	0.985	0.991	0.999	0.999
ProTrek structure-to-sequence	TP	25612	32882	32114	4441	4053	6426	6123	54129	212237
	FP	827	6872	8674	2146	581	309	41	27	23
	Precision	0.969	0.827	0.787	0.674	0.875	0.954	0.993	0.999	0.999
ProTrek sequence-to-structure	TP	15416	47609	43770	7602	4459	6307	5740	50521	208276
	FP	530	7098	9244	2054	296	167	54	35	22
	Precision	0.967	0.870	0.826	0.787	0.938	0.974	0.991	0.999	0.999
ProTrek sequence-to-sequence	TP	45903	56298	57458	12105	7727	6182	6321	56052	220580
	FP	801	7019	8773	2167	356	232	99	30	23
	Precision	0.983	0.889	0.868	0.848	0.956	0.964	0.985	0.999	0.999

Supplementary Table 13: ProTrek search precision using protein sequence/structure as input. Clearly, when the query protein and candidate proteins have very high TM-scores, all models achieve high true positives and low false positives. The main difference is that ProTrek demonstrates considerably superior performance in the TM-score range below 0.5.

Sequence-to-sequence				
Input: FFSPSPARKRHAPSPEPAVQGTGVAGVPEESGDAAAIPAKKAPAGQEEPGTPPSSPLSAEQLDRIQR NKAAALLRLAARNVPVGFGESWKKHLSGEFGKPYFIKLMGFVAEERKHYTVYPPPHQVFTWTQMCDI KDVKVVILGQDPAHGPNQAHGLCFSVQRPVPPPPSLENIYKELSTDIEDFVHPGHGDLSGWAKQGVL LLNAVLTVRAHQANSHKERGWEQFTDAVVSWLNQNSNGLVFLLWGSYAQKKGSAIDRKRHHVLQTAH PSPLSVYRGFFGCRHFSKTNELLQKSGKKPIDWKEL				
ID	**Sequence**	**Length**	**Sequence identity**	**ProTrek matching score**
3300046836|Ga0495477_0001733 |CDS|Ga0495477_0001733_89 30_9838|+|8930:9838	VTARRERYLEEMGLGPVWKLRARGRQDREPGVPDSEGGNRAPEKAAEIRREPRPGTPEAAVPS TAAPGPEAGGEDRVRLEPSWKARLFHEFSQPYMRDLREFLKKEISAKKVIYPRGSEYFAALDH TPFDKVKVVILGQDPYHGPRQAHGLCFSVRPGVEVPPSLQNIFIELKNDLGIEPPDHGHLTHW AEQGVLLLNATLSVEAGKAGSHQNKGWEKFTDAVIDVLNREREGLVFVLWGSYAQKKGAFIDT KKHLVLQGPHPSPLSAHRGFFGCRHFSKINSYLQAHRQAPIDWRLPKLNG	302	40.24%	43.78607537
3300010389|Ga0136549_1000130 2|CDS|Ga0136549_100013025 |-|10181:11125	MTQKRLDAFFRPTKPSVRAASVAIDGSGDSPAEVTSQTSEKRRKLASDDPQRERPPQATPQAL RANANKNLALVRQAVLACEKQGTAPALADLLVDSSWREALRGEFSKPYFLALERFVQSAWSSS MVFPPKACVFRAFNAVPLDQVRVVILGQDPYHDIGQAQGLSFSVPTGKKLPSSLRNIYKEISE DMGCSMPKENGCLEPWAHQGVLMLNAVLTVEAHKAASHSKKGWEEFTTHAVRVINKERSGVVF MLWGKYAQEKGSLIDVKRHHVLKAPHPSGLSAHRGFFGCKHFSQANALLTAQGDVPINWNVS	314	36.65%	41.92776629
3300010165|Ga0136203_100008 |CDS|Ga0136203_100008115 |+|169129:170037	MKRVTDYFATVPSKKPANARTISTPLTTKAGPPESNAVSAATPLTPRSSQSPSKPFKWDKDAW VAQLSPENRELLDLEIKTMDESWLAVLYKELTKPYFLSLKRFLTERRRNAQVFPPENDVYSWS RLTPLHNVKVLVLGQDPYHGINQAHGLAFSVKPPTRTPPSLLNIYKELRSCFPDDFEIPKHGF LEKWAQQGVLMLNACLTVDAHNANSHANKGWEQFTEQVVRAALNANDHVVIMAWGSPAAKRVD RVRPGSQHLVLRSVHPSPLSAARGWFGSAHFKKGNEWLEKMHREPVNWRL	302	34.00%	40.95163143
3300042934|Ga0456364_000000 1|CDS|Ga0456364_0000001_ 715595_716512|-|715595:71 6512	MKRVAAEVAVIKKAKITDYFKVKEQVKVVEKATIEEAVKKVSVDNTSSVGLNHSEFKQQFIKK LTADQLSLLQLEINTLEDSWFEAIHTELTKPYFLSLKRFLNQQRQSHKIFPPEEDVYSWSRLT PLHSVKCIILGQDPYHNDNQAHGLAFSVKEPTPPPPSLRNMYKALKLDYPQFEAPRTGDLTHW AQQGVLLLNACLTVRAHEANSHAKKGWEQFTESVLKVALAQQKELVLLLWGTPAQNRVKNMSL KKHHVLKCVHPSPLSASRGFFEAQHFKKCNDWLKEQGQTPIDWALEKGNVIDY	305	34.24%	40.78834125
ERZ1760321|ERZ1760321.34-NOD E-34-length-67626-cov-29.54265 9|CDS|ERZ1760321.34-NODE-3 4-length-67626-cov-29.542659_ 42|-|52802:53851	MAMTVNQKRKQTTIEDFFGAKKSATGTPSKKAKTSATFTSVRKDVVTKIEAKDVVKKVTASKD LANSDVKADFSEHLSTNLRDLLSLEVNTIDDSWFPHLMDEFKKPYFVKLKQFITKEQTDHTVF PPPNDIYSWSRLTPFDKVKVVIIGQDPYHNHNQAHGLAFSVKPPTPAPPSLKNIYKELKQEYP DFVQDNKVGDLTHWASQGVLLLNTSLSVRAHNANSHSKHGWETFTKRVVQLLLQDREADGKSL VFLLWGNNAIKLVESLLGSTSVGAGSKYPNIMVLKSVHPSPLSASRGFFGTNHFKKINDWLYN TRGEKMIDWSVVPGTSLREVEEANSREKPNSEEL	349	33.26%	40.40016404

Text-to-sequence			
Input: Uracil-DNA glycosylases remove uracil from DNA			
ID	**Sequence**	**Length**	**ProTrek matching score**
3300046836|Ga0495477_000173 3|CDS|Ga0495477_0001733_ 8930_9838|+|8930:9838	VTARRERYLEEMGLGPVWKLRARGRQDREPGVPDSEGGNRAPEKAAEIRREPRPGTPEAAVPS TAAPGPEAGGEDRVRLEPSWKARLFHEFSQPYMRDLREFLKKEISAKKVIYPRGSEYFAALDH TPFDKVKVVILGQDPYHGPRQAHGLCFSVRPGVEVPPSLQNIFIELKNDLGIEPPDHGHLTHW AEQGVLLLNATLSVEAGKAGSHQNKGWEKFTDAVIDVLNREREGLVFVLWGSYAQKKGAFIDT KKHLVLQGPHPSPLSAHRGFFGCRHFSKINSYLQAHRQAPIDWRLPKLNG	302	19.38698044
ERZ3455868|ERZ3455868.12977- NODE-12978-length-2261-cov-2.8 41342|CDS|ERZ3455868.12977 -NODE-12978-length-2261-cov-2. 841342_2|+|502:1464	MTLGSRVTLKHQHALVGTSSLASIAHSLLAPQAGHRAGVGMVNPIKPWELAAFIQQLGDRHGH PRQTCLIALTRRVGFFSVRTIVFRDVAMTLDDRIKLEPLWKQALREEFEKPYMGELREFLRQE HAAGKEIYPPGPLIFNALNSTPLVKVKVVILGQDPYHGPGQAHGLCFSVQRGVPAPPSLVNIY KELKRDLNIDIPSHGCLQSWADQGVLLLNTTMTVERANAASHAKKGWEHFTDRIIQVVSETQP TLVFLLWGAHAQGKQKLIDATKHLVLTSVHPSPLSAYRGFIGNGHFSRTNKFLEQNGLTPINW TLPAD	320	19.33582261
3300048834|Ga0498918_001045 0|CDS|Ga0498918_0010450_ 11002_11949|-|11002:11949	LFKTERKKERKTIMQSLPSLVGDSETPIIINSNDINSNNNTSNNTIGNDNVISIHLDIDITSI RPGKLPIPDDWSLERIIQERDPPFGWENVFRESVDELIHISHVLDRLGEYYPKKENLFAAFDK TPLENVRVILIGQDPYHQTKSHSSSPRAQGTSFSFARDDPVPGNSSLYNIFTEIHNSYPESFV MPQHGDLTSWTDQGVLLLNTCLTVRPGQAESHYINKRNIWHPFLIRVLNAIIHREQKNGTKPI FILLGGKAQELARELGEKVVILNASHPSPLSAHRGFFGSDVFKKANLWLQSFGQRPIDWNLYP	315	19.23660258
3300021963|Ga0222712_100007 95|CDS|Ga0222712_10000795 22|-|21482:22462	MQACRVFSNRLLYAQNFRAMSDKRKERSITSFFKPVNASVVDKSSYDGAKSEEPADKKTRVVN IMPAVAAVPVTTTDGDNSVENALPLMGYSSLLKNEQWRSILSGEFRKPYWLNLEKFLESEQRA GAKIFPSRENVFRAFDSCSMENIKVVIIGQDPYHDDGQAMGLCFSVPKGVQVPSSLLNIYKEL SNDLGCKVPTHGDLDKWAKQGVLLLNASLTVRAHAANSHSKKGWEMFTDAAIREVSKKRTGLV FLLWGKNAQDKEALISKANGHVVLKCPHPSGLSAHRGFFGSKHFSSANEAITKKGGTPIDWQI EGPYRLLPPKP	326	19.23409895
3300022240|Ga0236259_131145 |CDS|Ga0236259_131145126 |+|126443:127366	MEVTAFHRNLIRSSLWPYSSPYTVTSFQRTAVSRHPALCSPDFPPPVHTGGDCLVGFGAILPC LAEHPIMHKSLNELNDMTTQEKPGQLDPSWQAVLGAEFDKPYMQSLKAFLQQEKTAGKTIFPP GPLIFNAFNHTPFDQVRVVIIGQDPYHGPGQAHGLSFSVQAGVALPPSLMNIFKEIQSDLGIR MSGSGDLTPWADQGVLLLNATLTVQMANAGSHQNRGWETFTDAAIAALNAQREGLVFVLWGSY AQKKGAMIDTRKHLVLKSVHPSPLSAHRGFFGTHQFSKINEYLVSRGEAPINWQL	307	19.03161063

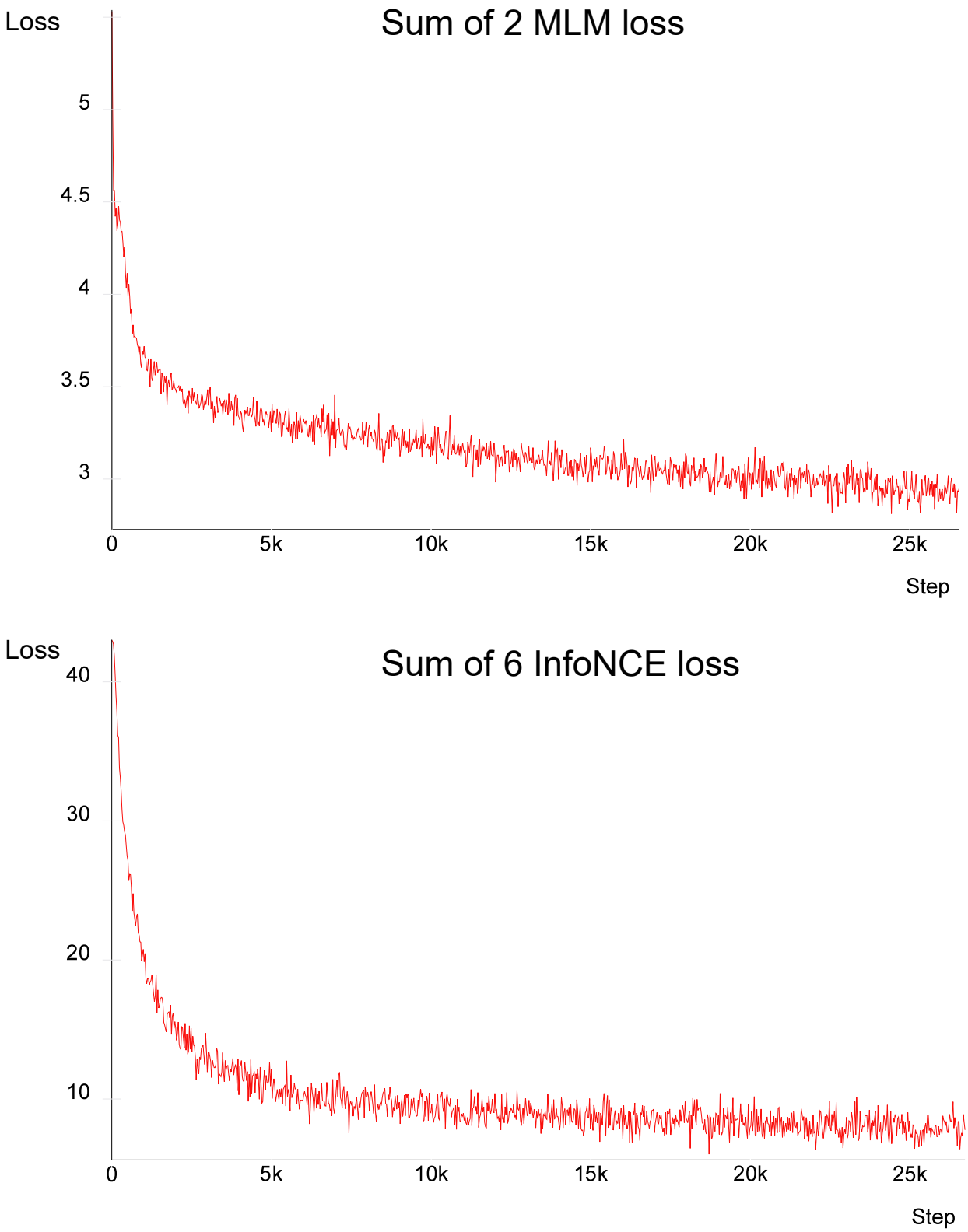
Supplementary Table 14: We searched for proteins using ProTrek's sequence-to-sequence and text-to-sequence searches. We queried the OMG_prot50 database using both the human UDG protein sequence and the functional description "Uracil-DNA glycosylases remove uracil from DNA", retrieving the top five matches from each approach independently.

Base editor	Protein sequence
TSBE-V1	PKKKRKVDKKYSIGLAIGTNSVGWAVITDEYKVPSKKFKVLGNTDRHSIKKNLIGALLFDSGETAEATRLKRTARRRYTRRKNRICYLQEIFSNEMAKVDD SFFHRLEESFLVEEDKKHERHPIFGNIVDEVAYHEKYPTIYHLRKKLVDSTDKADLRLIYLALAHMIKFRGHFLIEGDLNPDNSDVDKLFIQLVQTYNQLF EENPINASGVDAKAILSARLSKSRRLENLIAQLPGEKKNGLFGNLIALSLGLTPNFKSNFDLAEDAKLQLSKDTYDDDLDNLLAQIGDQYADLFLAAKNLSD AILLSDILRVNTEITKAPLSASMIKRYDEHHQDLTLLKALVRQQLPEKYKEIFFDQSKNGYAGYIDGGASQEEFYKFIKPILEKMDGTEELLVKLNREDLLRK QRTFDNGSIPHQIHLGELHAILRRQEDFYPFLKDNREKIEKILTFRIPYYVGPLARGNSRFAWMTRKSEETITPWNFEEVVDKGASAQSFIERMTNFDKNL PNEKVLPKHSLLYEYFTVYNELTKVKYVTEGMRKPAFLSGEQKKAIVDLLFKTNRKVTVKQLKEDYFKKIECFDSVEISGVEDRFNASLGTYHDLLKIIKDK DFLDNEENEDILEDIVLTLTLFEDREMIEERLKTYAHLFDDKVMKQLKRRRYTGWGRLSRKLINGIRDKQSGKTILDFLKSDGFANRNFMQLIHDDSLTFK EDIQKAQVSGQGDSLHEHIANLAGSPAIKKGILQTVKVVDELVKVMGRHKPENIVIEMARENQTTQKGQKNSRERMKRIEEGIKELGSQILKEHPVENTQ LQNEKLYLYYLQNGRDMYVDQELDINRLSDYDVDAIVPQSFLKDDSIDNKVLTRSDKNRGKSDNVPSEEVVKKMKNYWRQLLNAKLITQRKFDNLTKAE RGGLSELDKAGFIKRQLVETRQITKHVAQILDSRMNTKYDENDKLIREVKVITLKSKLVSDFRKDFQFYKVREINNYHHAHDAYLNAVVGTALIKKYPKLE SEFVYGDYKVYDVRKMIAKSEQEIGKATAKYFFYSNIMNFFKGSSGSVTARRERYLEEMGLGPVWKLRARGRQDREPGVPDSEGGNRAPEKAAEIRREPR PGTPEAAVPSTAAPGPEAGGEDRVRLEPSWKARLFHEFSQPYMRDLREFLKKEISAKKVIYPRGSEYFAALDHTPFDKVKVVILGQDPAHGPRQAHGLCFS VRPGVEVPPSLQNIFIELKNDLGIEPPDHGHLTHWAEQGVLLLNATLSVEAGKAGSHQNKGWEKFTDAVIDVLNREREGLVFVLWGSYAQKKGAFIDTKK HLVLQGPHPSPLSAHRGFFGCRHFSKINSYLQAHRQAPIDWRLPKLNGGSSGSETNGETGEIVWDKGRDFATVRKVLSMPQVNIVKKTEVQTGGFSKESI LPKRNSDKLIARKKDWDPKKYGGFDSPTVAYSVLVVAKVEKGKSKKLKSVKELLGITIMERSSFEKNPIDFLEAKGYKEVKKDLIIKLPKYSLFELENGRKR MLASAGELQKGNELALPSKYVNFLYLASHYEKLKGSPEDNEQKQLFVEQHKHYLDEIIEQISEFSKRVILADANLDKVLSAYNKHRDKPIREQAENIIHLFT LTNLGAPAAFKYFDTTIDRKRYTSTKEVLDATLIHQSITGLYETRIDLSQLGGDPKKKRKV
TSBE3 EK	PKKKRKVDKKYSIGLAIGTNSVGWAVITDEYKVPSKKFKVLGNTDRHSIKKNLIGALLFDSGETAEATRLKRTARRRYTRRKNRICYLQEIFSNEMAKVDD SFFHRLEESFLVEEDKKHERHPIFGNIVDEVAYHEKYPTIYHLRKKLVDSTDKADLRLIYLALAHMIKFRGHFLIEGDLNPDNSDVDKLFIQLVQTYNQLF EENPINASGVDAKAILSARLSKSRRLENLIAQLPGEKKNGLFGNLIALSLGLTPNFKSNFDLAEDAKLQLSKDTYDDDLDNLLAQIGDQYADLFLAAKNLSD AILLSDILRVNTEITKAPLSASMIKRYDEHHQDLTLLKALVRQQLPEKYKEIFFDQSKNGYAGYIDGGASQEEFYKFIKPILEKMDGTEELLVKLNREDLLRK QRTFDNGSIPHQIHLGELHAILRRQEDFYPFLKDNREKIEKILTFRIPYYVGPLARGNSRFAWMTRKSEETITPWNFEEVVDKGASAQSFIERMTNFDKNL PNEKVLPKHSLLYEYFTVYNELTKVKYVTEGMRKPAFLSGEQKKAIVDLLFKTNRKVTVKQLKEDYFKKIECFDSVEISGVEDRFNASLGTYHDLLKIIKDK DFLDNEENEDILEDIVLTLTLFEDREMIEERLKTYAHLFDDKVMKQLKRRRYTGWGRLSRKLINGIRDKQSGKTILDFLKSDGFANRNFMQLIHDDSLTFK EDIQKAQVSGQGDSLHEHIANLAGSPAIKKGILQTVKVVDELVKVMGRHKPENIVIEMARENQTTQKGQKNSRERMKRIEEGIKELGSQILKEHPVENTQ LQNEKLYLYYLQNGRDMYVDQELDINRLSDYDVDAIVPQSFLKDDSIDNKVLTRSDKNRGKSDNVPSEEVVKKMKNYWRQLLNAKLITQRKFDNLTKAE RGGLSELDKAGFIKRQLVETRQITKHVAQILDSRMNTKYDENDKLIREVKVITLKSKLVSDFRKDFQFYKVREINNYHHAHDAYLNAVVGTALIKKYPKLE SEFVYGDYKVYDVRKMIAKSEQEIGKATAKYFFYSNIMNFFKGSSGSVTARRERYLEEMGLGPVWKLRARGRQDREPGVPDSEGGNRAPEKAAEIRREPR PGTPEAAVPSTAAPGPEAGGEDRVRLEPSWKARLFHEFSQPYMRDLREFLKKEISAKKVIYPRGSEYFAALDHTPFDKVKVVILGQDPAHGPRQAHGLCFS VRPGVEVPPSLQNIFIELKNDLGIEPPDHGHLTHWAEQGVLLLNATLSVEAGKAGSHQNKGWEKFTDAVIDVLNREREGLVFVLWGSYAQKKGAFIDTKK HLVLQGPHPSPLSAHRGFFGCRHFSKINSYLQAHRQAPIDWRLPKLNGGSSGSETNGETGEIVWDKGRDFATVRKVLSMPQVNIVKKTEVQTGGFSKESI LPKRNSDKLIARKKDWDPKKYGGFDSPTVAYSVLVVAKVEKGKSKKLKSVKELLGITIMERSSFEKNPIDFLEAKGYKEVKKDLIIKLPKYSLFELENGRKR MLASAGELQKGNELALPSKYVNFLYLASHYEKLKGSPEDNEQKQLFVEQHKHYLDEIIEQISEFSKRVILADANLDKVLSAYNKHRDKPIREQAENIIHLFT LTNLGAPAAFKYFDTTIDRKRYTSTKEVLDATLIHQSITGLYETRIDLSQLGGDPKKKRKV
gTBE	PKKKRKVDKKYSIGLAIGTNSVGWAVITDEYKVPSKKFKVLGNTDRHSIKKNLIGALLFDSGETAEATRLKRTARRRYTRRKNRICYLQEIFSNEMAKVDD SFFHRLEESFLVEEDKKHERHPIFGNIVDEVAYHEKYPTIYHLRKKLVDSTDKADLRLIYLALAHMIKFRGHFLIEGDLNPDNSDVDKLFIQLVQTYNQLF EENPINASGVDAKAILSARLSKSRRLENLIAQLPGEKKNGLFGNLIALSLGLTPNFKSNFDLAEDAKLQLSKDTYDDDLDNLLAQIGDQYADLFLAAKNLSD AILLSDILRVNTEITKAPLSASMIKRYDEHHQDLTLLKALVRQQLPEKYKEIFFDQSKNGYAGYIDGGASQEEFYKFIKPILEKMDGTEELLVKLNREDLLRK QRTFDNGSIPHQIHLGELHAILRRQEDFYPFLKDNREKIEKILTFRIPYYVGPLARGNSRFAWMTRKSEETITPWNFEEVVDKGASAQSFIERMTNFDKNL PNEKVLPKHSLLYEYFTVYNELTKVKYVTEGMRKPAFLSGEQKKAIVDLLFKTNRKVTVKQLKEDYFKKIECFDSVEISGVEDRFNASLGTYHDLLKIIKDK DFLDNEENEDILEDIVLTLTLFEDREMIEERLKTYAHLFDDKVMKQLKRRRYTGWGRLSRKLINGIRDKQSGKTILDFLKSDGFANRNFMQLIHDDSLTFK EDIQKAQVSGQGDSLHEHIANLAGSPAIKKGILQTVKVVDELVKVMGRHKPENIVIEMARENQTTQKGQKNSRERMKRIEEGIKELGSQILKEHPVENTQ LQNEKLYLYYLQNGRDMYVDQELDINRLSDYDVDAIVPQSFLKDDSIDNKVLTRSDKNRGKSDNVPSEEVVKKMKNYWRQLLNAKLITQRKFDNLTKAE RGGLSELDKAGFIKRQLVETRQITKHVAQILDSRMNTKYDENDKLIREVKVITLKSKLVSDFRKDFQFYKVREINNYHHAHDAYLNAVVGTALIKKYPKLE SEFVYGDYKVYDVRKMIAKSEQEIGKATAKYFFYSNIMNFFKGSSGSVTARRERYLEEMGLGPVWKLRARGRQDREPGVPDSEGGNRAPEKAAEIRREPR PGTPEAAVPSTAAPGPEAGGEDRVRLEPSWKARLFHEFSQPYMRDLREFLKKEISAKKVIYPRGSEYFAALDHTPFDKVKVVILGQDPAHGPRQAHGLCFS VRPGVEVPPSLQNIFIELKNDLGIEPPDHGHLTHWAEQGVLLLNATLSVEAGKAGSHQNKGWEKFTDAVIDVLNREREGLVFVLWGSYAQKKGAFIDTKK HLVLQGPHPSPLSAHRGFFGCRHFSKINSYLQAHRQAPIDWRLPKLNGGSSGSETNGETGEIVWDKGRDFATVRKVLSMPQVNIVKKTEVQTGGFSKESI LPKRNSDKLIARKKDWDPKKYGGFDSPTVAYSVLVVAKVEKGKSKKLKSVKELLGITIMERSSFEKNPIDFLEAKGYKEVKKDLIIKLPKYSLFELENGRKR MLASAGELQKGNELALPSKYVNFLYLASHYEKLKGSPEDNEQKQLFVEQHKHYLDEIIEQISEFSKRVILADANLDKVLSAYNKHRDKPIREQAENIIHLFT LTNLGAPAAFKYFDTTIDRKRYTSTKEVLDATLIHQSITGLYETRIDLSQLGGDPKKKRKV
TSBE-eGFP	PKKKRKVDKKYSIGLAIGTNSVGWAVITDEYKVPSKKFKVLGNTDRHSIKKNLIGALLFDSGETAEATRLKRTARRRYTRRKNRICYLQEIFSNEMAKVDD SFFHRLEESFLVEEDKKHERHPIFGNIVDEVAYHEKYPTIYHLRKKLVDSTDKADLRLIYLALAHMIKFRGHFLIEGDLNPDNSDVDKLFIQLVQTYNQLF EENPINASGVDAKAILSARLSKSRRLENLIAQLPGEKKNGLFGNLIALSLGLTPNFKSNFDLAEDAKLQLSKDTYDDDLDNLLAQIGDQYADLFLAAKNLSD AILLSDILRVNTEITKAPLSASMIKRYDEHHQDLTLLKALVRQQLPEKYKEIFFDQSKNGYAGYIDGGASQEEFYKFIKPILEKMDGTEELLVKLNREDLLRK QRTFDNGSIPHQIHLGELHAILRRQEDFYPFLKDNREKIEKILTFRIPYYVGPLARGNSRFAWMTRKSEETITPWNFEEVVDKGASAQSFIERMTNFDKNL PNEKVLPKHSLLYEYFTVYNELTKVKYVTEGMRKPAFLSGEQKKAIVDLLFKTNRKVTVKQLKEDYFKKIECFDSVEISGVEDRFNASLGTYHDLLKIIKDK DFLDNEENEDILEDIVLTLTLFEDREMIEERLKTYAHLFDDKVMKQLKRRRYTGWGRLSRKLINGIRDKQSGKTILDFLKSDGFANRNFMQLIHDDSLTFK EDIQKAQVSGQGDSLHEHIANLAGSPAIKKGILQTVKVVDELVKVMGRHKPENIVIEMARENQTTQKGQKNSRERMKRIEEGIKELGSQILKEHPVENTQ LQNEKLYLYYLQNGRDMYVDQELDINRLSDYDVDAIVPQSFLKDDSIDNKVLTRSDKNRGKSDNVPSEEVVKKMKNYWRQLLNAKLITQRKFDNLTKAE RGGLSELDKAGFIKRQLVETRQITKHVAQILDSRMNTKYDENDKLIREVKVITLKSKLVSDFRKDFQFYKVREINNYHHAHDAYLNAVVGTALIKKYPKLE SEFVYGDYKVYDVRKMIAKSEQEIGKATAKYFFYSNIMNFFKGSSGSVTARRERYLEEMGLGPVWKLRARGRQDREPGVPDSEGGNRAPEKAAEIRREPR PGTPEAAVPSTAAPGPEAGGEDRVRLEPSWKARLFHEFSQPYMRDLREFLKKEISAKKVIYPRGSEYFAALDHTPFDKVKVVILGQDPAHGPRQAHGLCFS VRPGVEVPPSLQNIFIELKNDLGIEPPDHGHLTHWAEQGVLLLNATLSVEAGKAGSHQNKGWEKFTDAVIDVLNREREGLVFVLWGSYAQKKGAFIDTKK HLVLQGPHPSPLSAHRGFFGCRHFSKINSYLQAHRQAPIDWRLPKLNGGSSGSETNGETGEIVWDKGRDFATVRKVLSMPQVNIVKKTEVQTGGFSKESI LPKRNSDKLIARKKDWDPKKYGGFDSPTVAYSVLVVAKVEKGKSKKLKSVKELLGITIMERSSFEKNPIDFLEAKGYKEVKKDLIIKLPKYSLFELENGRKR MLASAGELQKGNELALPSKYVNFLYLASHYEKLKGSPEDNEQKQLFVEQHKHYLDEIIEQISEFSKRVILADANLDKVLSAYNKHRDKPIREQAENIIHLFT LTNLGAPAAFKYFDTTIDRKRYTSTKEVLDATLIHQSITGLYETRIDLSQLGGDPKKKRKV

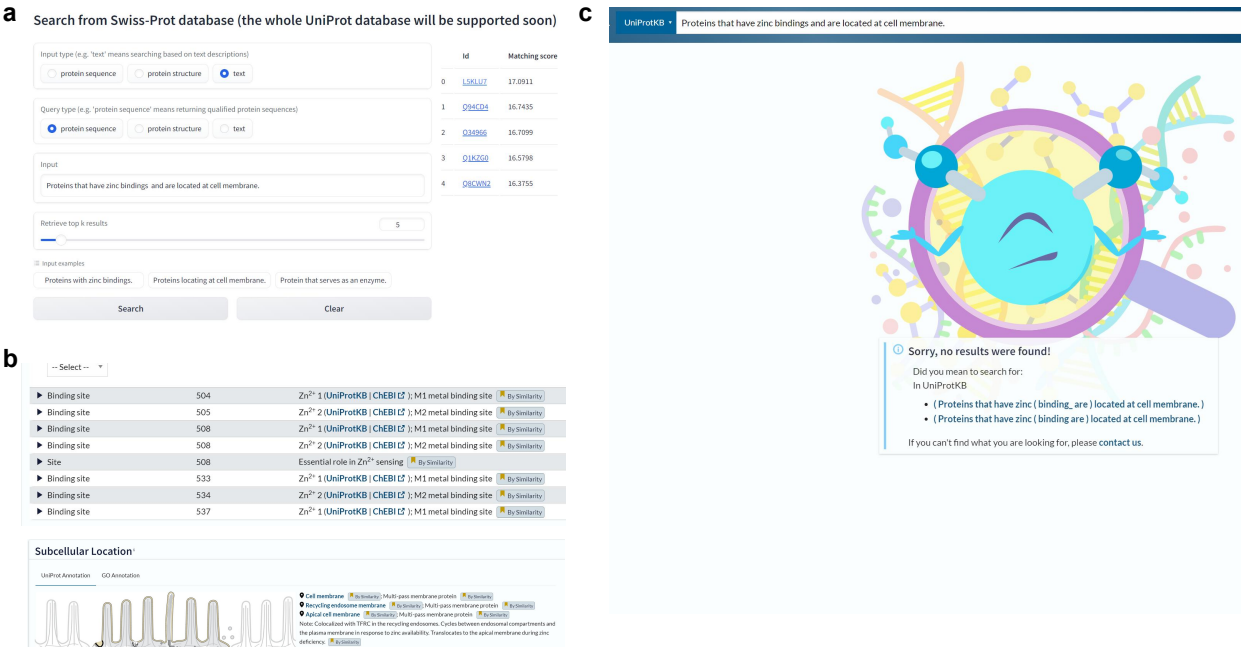
Supplementary Table 15: All plasmid protein sequences used in this study.

sgRNA	Amplicon	Spacer	Primer-F	Primer-R
Dicer-1	tgattcaccatgGAATTGA AACATTATAAAGATCATTT CTGCTACCACTATTTAAAC GTTATTACCCACTTGAATT GAAAAAAGATGAAAGGACA CAAAGTGAACAGACGATAA CTTTATTGGAGATTTACTT GGCTACAATTATTACAAAA AAAACAACTAAAGGTAATT GTGATCCATAAGGTGCAGA CTGCAAATTTCTGCAGCAT GATTACTGTATGAATGAAT	AACTTTATTGGAGATTTACTT GG	ggagtgagtacggtgtgcAT TCATTCATACAGTAATCAT GCTG	gagttggatgctggatggtg attcaccatgGAATTGAAAC
VEGFA	ttctcccctgggaagcatc cctggacacttcccaaagg accccagtcactccagcct gttggctgccgctcacttt gatgtctgcaggccagatg agggctccagatggcacat tgtcagagggacacactgt ggcccctgtgcccagccct gggctctctgtacatgaag caactccagtcccaaatat gtagctgtttgggaggtca gaaa	gatgtctgcaggccagatg aggg	ggagtgagtacggtgtg cttctcccctgggaagc atcc	gagttggatgctggatggtt tctgacctcccaaacagc
#7	agacagggatcccagggaa acgcccatgcaattagtct atttctgctgcaagtaagc atgcatttgtaggcttgat gctttttttctgcttctcc agcCCTGGCCTGGGTCAAT CCTTGGGgcccagactgag cacgtgatggcagaggaaa ggaagccctgcttcctcca gagggcgtcgcaggacagc ttttcctagacaggggcta gtatgtgcagct	CCTGGCCTGGGTCAATCC TTGGG	ggagtgagtacggtgtg cagacagggatcccagg gaaa	gagttggatgctggatggag ctgcacatactagcccct
#14	gagacagggatcccaggga aacgcccatgcaattagtc tatttctgctgcaagtaag catgcatttgtaggcttga tgctttttttctgcttctc cagccCTGGCCTGGGTCAA TCCTTGGGGcccagactga gcacgtgatggcagaggaa aggaagccctgcttcctcc agagggcgtcgcaggacag cttttcctagacaggggct agtatg	CTGGCCTGGGTCAATCCTT GGGG	ggagtgagtacggtgtg cgagacagggatcccag ggaa	gagttggatgctggatggca tactagccctgtctagg
#18	gctcctgccaacctttcaa aagaggaagctgtctccgc tcccctgcccacgcaccag aagccctaggcctggcctt gggagcttcctgtcacctc acaGAGCTCACTGAACGCT GGCATGGctctagtgcttt ccaggggcacgtactcccc cgagctcctgacagcacaa tccacgtgtcacgcatgac ccctcatacagttgcgcac cataccattc	GAGCTCACTGAACGCTGG CATGG	ggagtgagtacggtgtg cgctcctgccaacctttt caaa	gagttggatgctggatggga atggtatggtgcgcaact

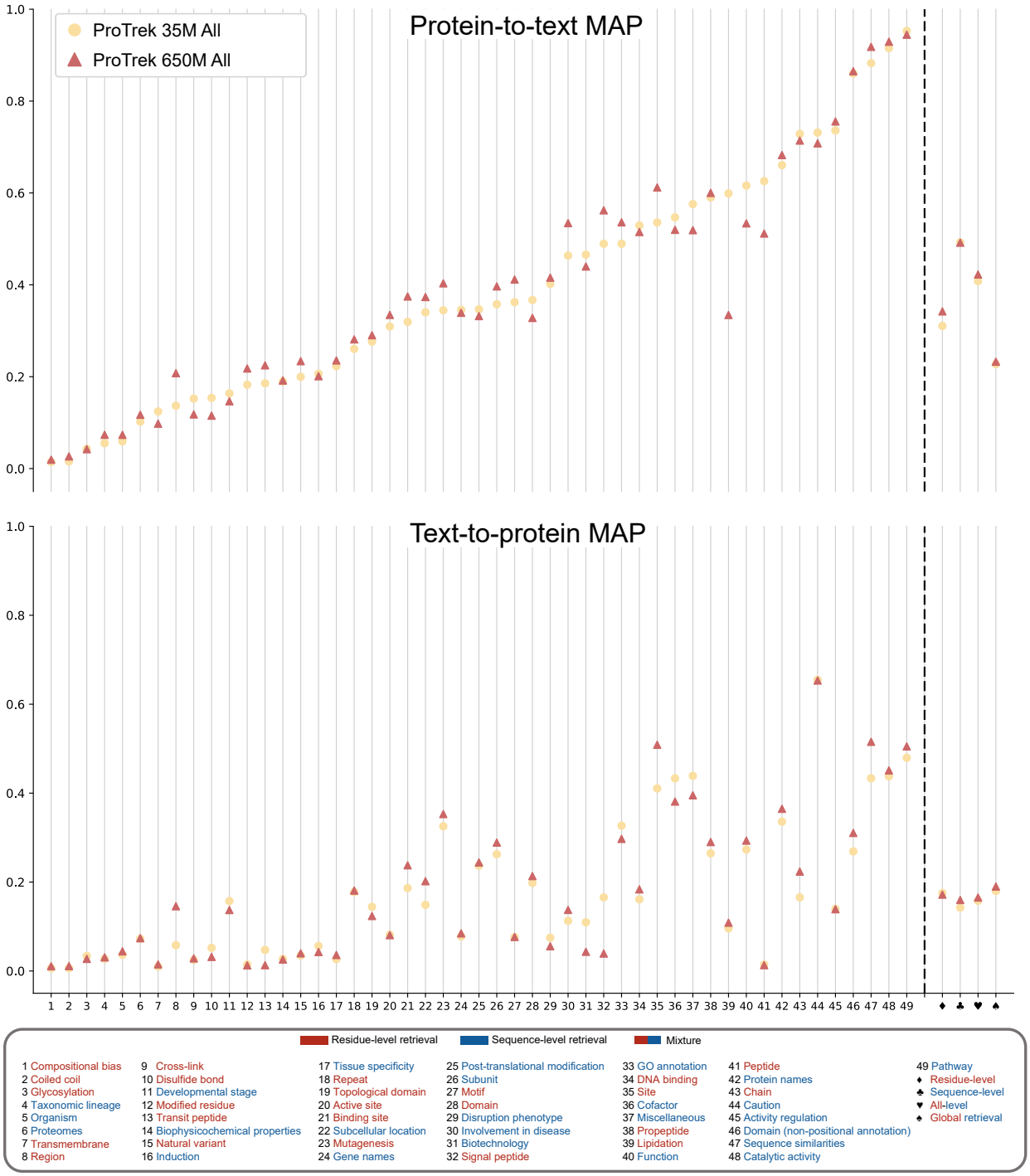
Supplementary Table 16: All primers used in this study.



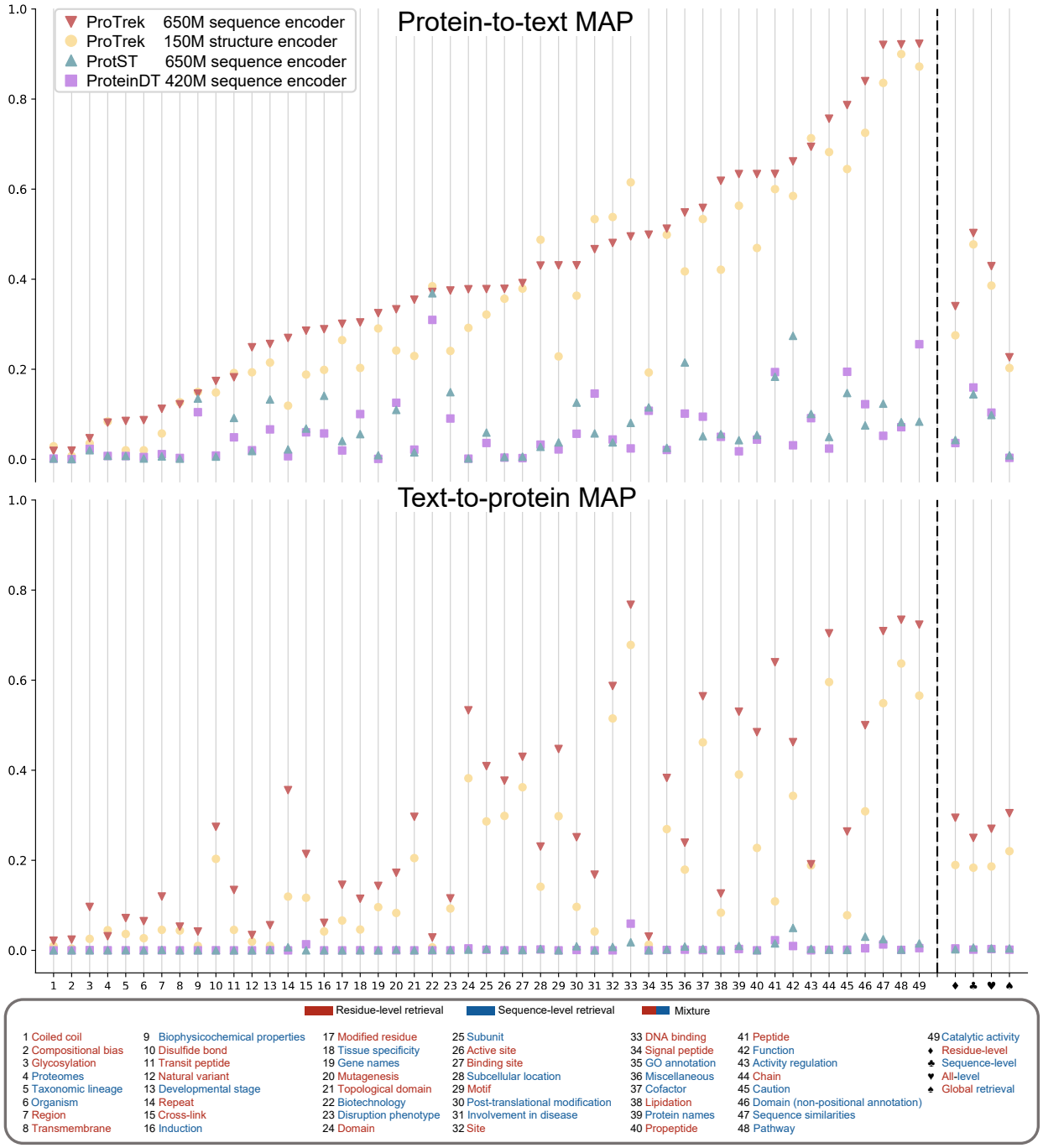
Supplementary Fig. 1: The loss trajectories of the sum of two MLM losses and the sum of six InfoNCE losses during the pre-training of the ProTrek 35M model. The results show that the two loss functions converge to the same order of magnitude after approximately 2,500 training steps.



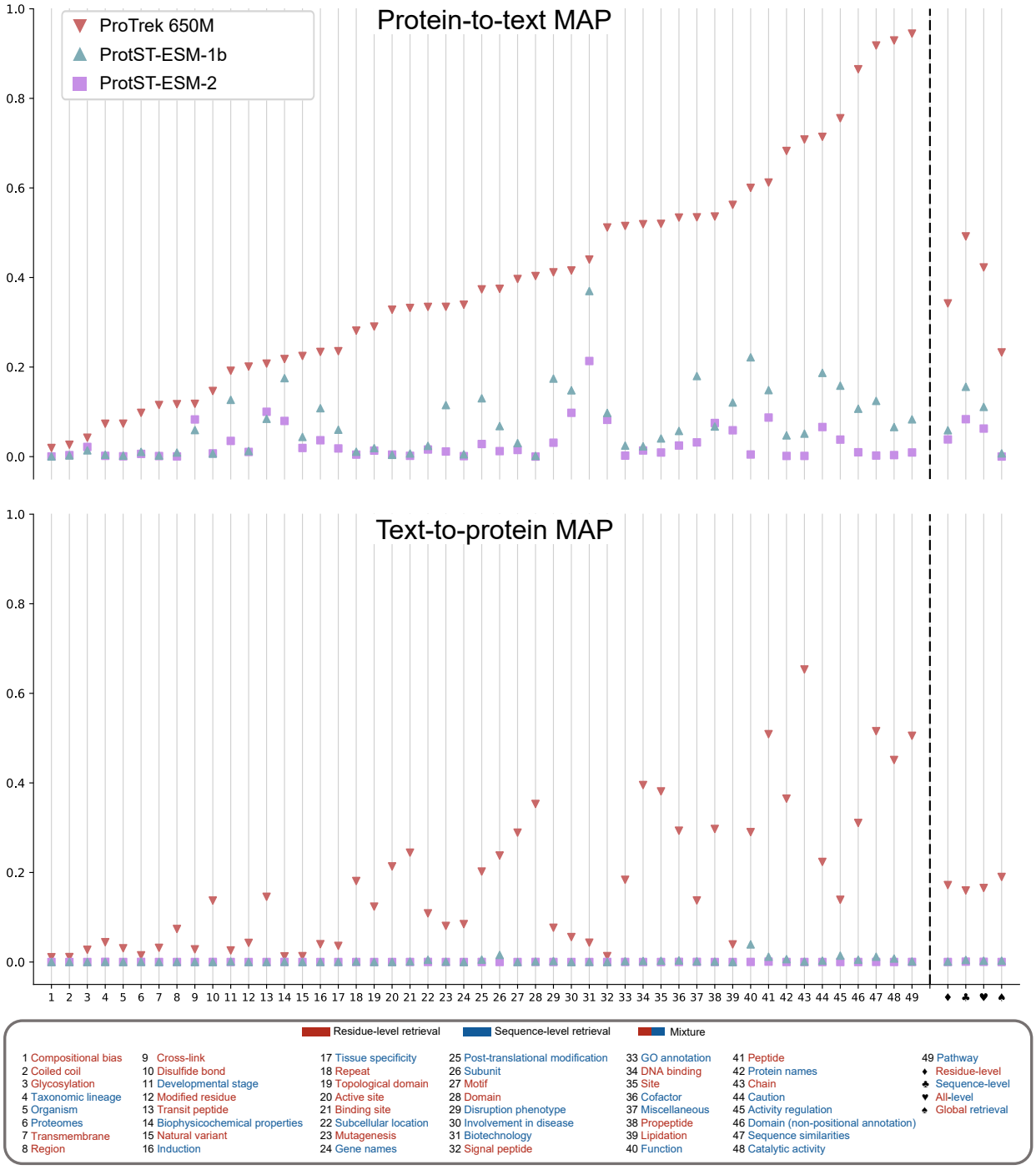
Supplementary Fig. 2: A case study using ProTrek to identify proteins with a long functional description. a, Using ProTrek demo to search proteins with the long functional description "Proteins that have zinc bindings and are located at cell membrane." **b,** "L5KLU7," a human-reviewed protein, is returned as the top-ranked result by ProTrek, matching the corresponding description exactly. **c,** Using the same description to search against the UniProt database. The UniProt database only supports keyword searches and did not return any results for the long natural sentence.



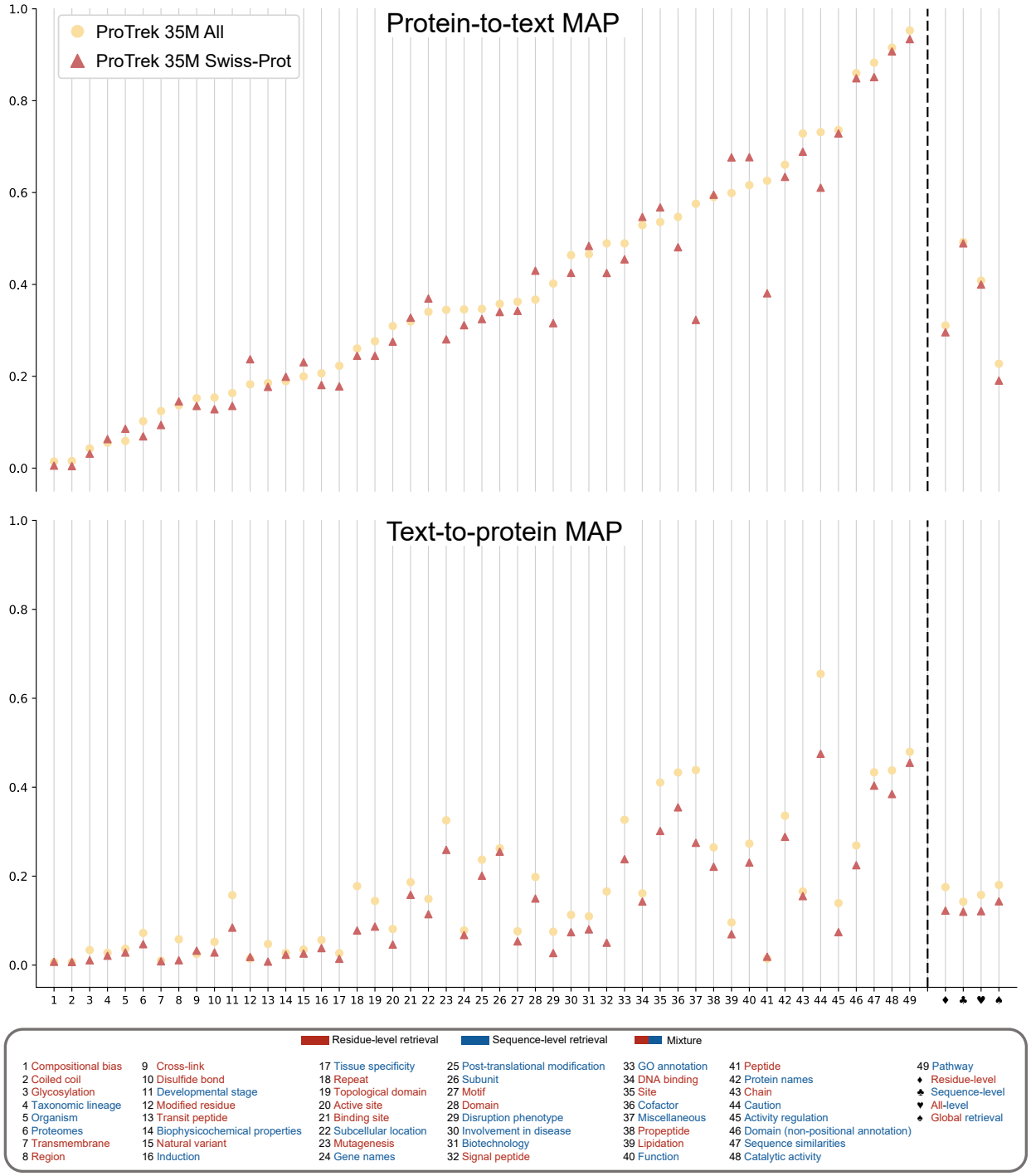
Supplementary Fig. 3: Performance comparison of the ProTrek 35M and ProTrek 650M.



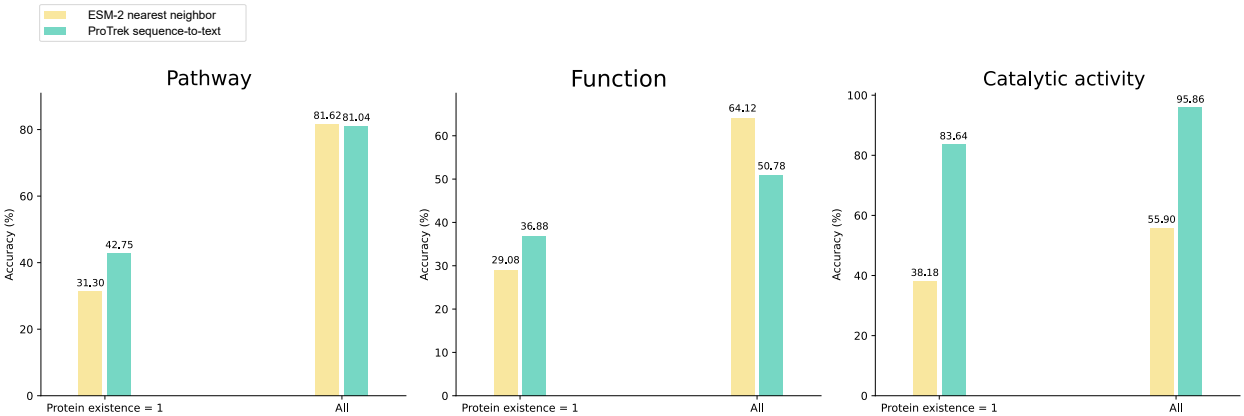
Supplementary Fig. 4: Retrieval results in the filtered test set. The "filtered test set" was created by further excluding proteins from both the test set and negative samples (i.e., the 100,000 randomly sampled proteins from UniProt/TrEMBL) that share more than 50% sequence identity with proteins in the 25 million noisy training examples. The number of proteins from Swiss-Prot in test set decreases from 4000 to 2058).The number of negative samples decreases from 100,000 to 63,000.



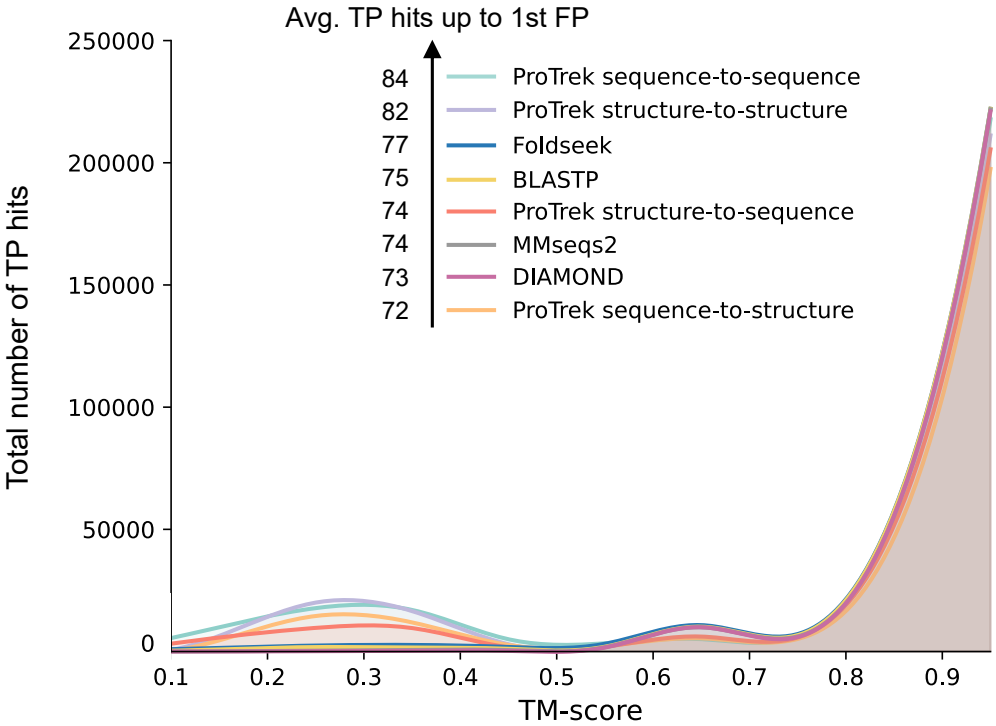
Supplementary Fig. 5: Comparison in the protein-text retrieval tasks between ProtST-ESM-1b and ProtST-ESM-2.



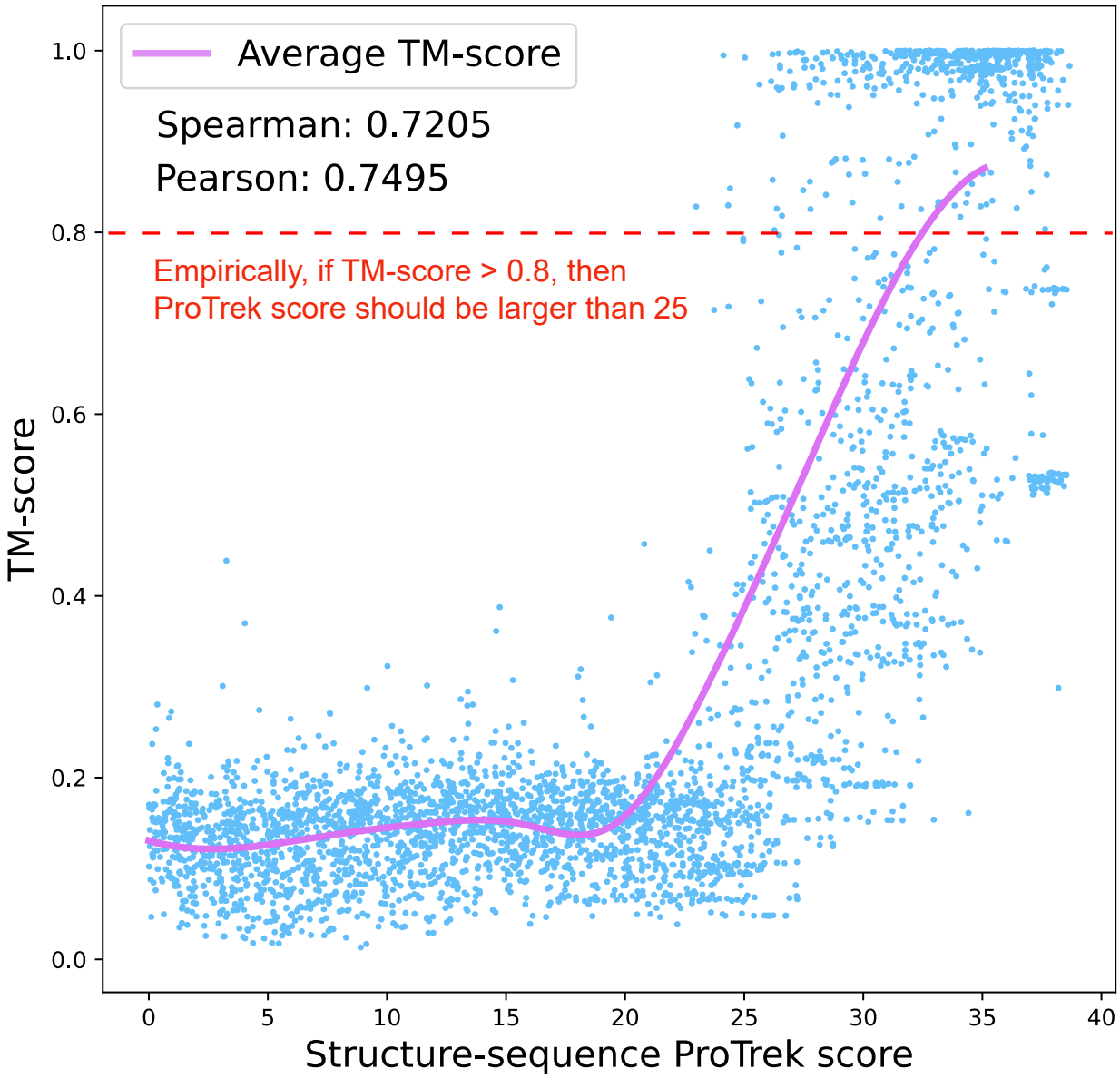
Supplementary Fig. 6: Comparison of protein-text retrieval between models trained on different datasets. "ProTrek 35M Swiss-Prot" was trained on Swiss-Prot data only and "ProTrek 35M All" was trained on a mixture of Swiss-Prot and TrEMBL50 data.



Supplementary Fig. 7: Comparison between ProTrek (sequence-to-text) and ESM-2 (nearest neighbor, NN) in the Pathway, Catalytic activity and Function prediction tasks with two experimental conditions: "Protein existence = 1" and "All". The "Protein existence = 1" condition exclusively includes testing set proteins classified as "Experimental evidence at protein level" (UniProt PE1) and excludes annotations containing the phrase "By similarity." This ensures that all ground-truth annotations used for evaluation are derived solely from clear experimental evidence, rather than homology-based inferences. The "All" condition uses the original testing set for a comparison purpose. For the nearest neighbor paradigm, the function predicted for a test protein is assigned using the description derived from its nearest neighbor which is calculated based on the cosine similarity in the ESM-2 embedding space. For ProTrek protein-to-text search, we only consider the top-1 searched results for a fair comparison. The use of a test set restricted to proteins with 'Protein Existence = 1' (PE1) as a more rigorous gold standard is primarily to mitigate a potential evaluation bias that arises when directly comparing ProTrek's protein-to-text search with a nearest-neighbor approach. The fundamental issue stems from the fact that approximately 80% of protein annotations in Swiss-Prot are not derived from clear experimental evidence but are instead inferred using methods such as ISS (inferred from sequence or structural similarity) with human review. This annotation methodology is conceptually aligned with the nearest neighbor approach. Given that the annotations for non-PE1 proteins in Swiss-Prot are not 100% accurate (see PMCID: PMC1892569), using a test set that includes these proteins creates an unfair setting for this evaluation scenario. As illustrated in this figure, ProTrek's protein-to-text search demonstrates a clear advantage on the PE1 test set. In contrast, on the "All" set (which includes a large volume of non-PE1 proteins), the nearest neighbor methods may appear better results (e.g. on the "Pathway" and "Function" prediction task). However, this better performance does not necessarily indicate the superiority of the NN method. Instead, it likely reflects the fact that the NN approach effectively simulates the inherent annotation process of Swiss-Prot itself, including its potential biases.



Supplementary Fig. 8: Searching proteins with similar functions using protein sequence/structure as input. TP (true-positive): Matches sharing ≥1 GO term. FP: Matches sharing no GO terms. Average TP hits up to the first FP are counted.



Supplementary Fig. 9: The correlation between the Structure-sequence ProTrek score and the TM-score. The result shows ProTrek's potential in accurately assessing the compatibility between protein structure and sequence.

References

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